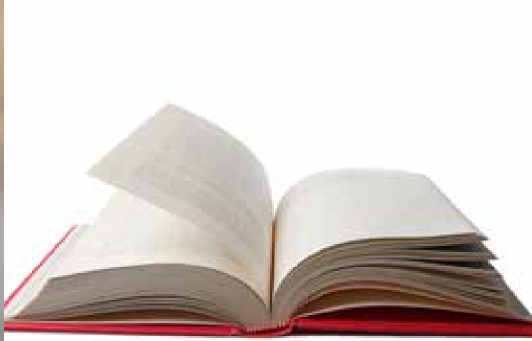




STRAND

Matter and Materials











6 Compounds

Key questions

- What is a compound?
- How is a compound different from an element?
- How is a mixture of elements different from a compound?
- What does the position of an element on the Periodic Table of elements tell us about its properties?
- Where do we find metals, non-metals and semi-metals on the Periodic Table?
- What are the vertical columns of the Periodic Table of elements called?
- What are the horizontal rows of the Periodic Table of elements called?
- What do elements belonging to the same 'group' of the Periodic Table of elements have in common?
- What additional information about an element can we find on the Periodic Table?
- What does the formula of a compound tell us about it?

Revision

- In a compound, atoms of two or more different kinds are chemically bonded in a fixed ratio.
- The atoms that make up a molecule are held together by special attractions called chemical bonds.
- Compounds can be formed and broken down in chemical reactions.
- A chemical reaction in which a compound is broken down into simpler compounds and even elements is called a decomposition reaction.
- Compounds cannot be separated by physical processes, but they can be separated into their elements (or simpler compounds) by chemical processes.

Keywords

- Periodic Table
- symbol (or element symbol)
- atomic number
- unique

6.1 The Periodic Table of elements

We first learnt about the Periodic Table in Grade 7. Here is a summary of what we already know:

1. All the elements that are known can be arranged on a table called the Periodic Table.
2. The discoveries of many scientists over many years contributed to the information in the Periodic Table, but the version of the table that we use today was originally proposed by Dmitri Mendeleev in the 1800s.
3. Each element has a fixed position on the Periodic Table. The elements are arranged in order of increasing atomic numbers, with the lightest element (hydrogen: H) in the top left-hand corner.
4. An element's position on the Periodic Table tells us whether it is a metal, a non-metal, or a semi-metal.
   a) metals are found on the left-hand side of the table;
   b) non-metals are found on the far right-hand side of the table; and
   c) semi-metals are found in the region between the metals and non-metals.
5. An element can be identified in 3 different ways:
   a) each element has a unique name;

b) each element has a unique chemical symbol; and
c) each element has a unique atomic number.

6. Metals are usually shiny, ductile, and malleable. Most are solids at room temperature and have high melting and boiling points.
7. Non-metals can be solids, liquids or gases at room temperature. They have a great variety of properties that usually depend on the state they are in.
8. The semi-metals are all solids at room temperature. They usually have a combination of metallic and non-metallic properties.

Take note

There is a large version of the Periodic Table printed on the inside cover of your workbooks for you to refer to easily.

We learnt about the origins of the Periodic Table in Grade 7. Let's also revise what we learnt then, so that we have a firm foundation for our new learning.

The Periodic Table is basically a chart that scientists use to list the known elements. The table consists of individual tiles for each of the elements. What information can we find on the Periodic Table? That is what the next section is all about.

What information can we find in the Periodic Table?

The information that most commonly appears on each tile of the Periodic Table is the following:

- The chemical symbol; and
- The atomic number
- The mass number

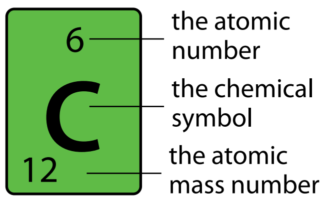
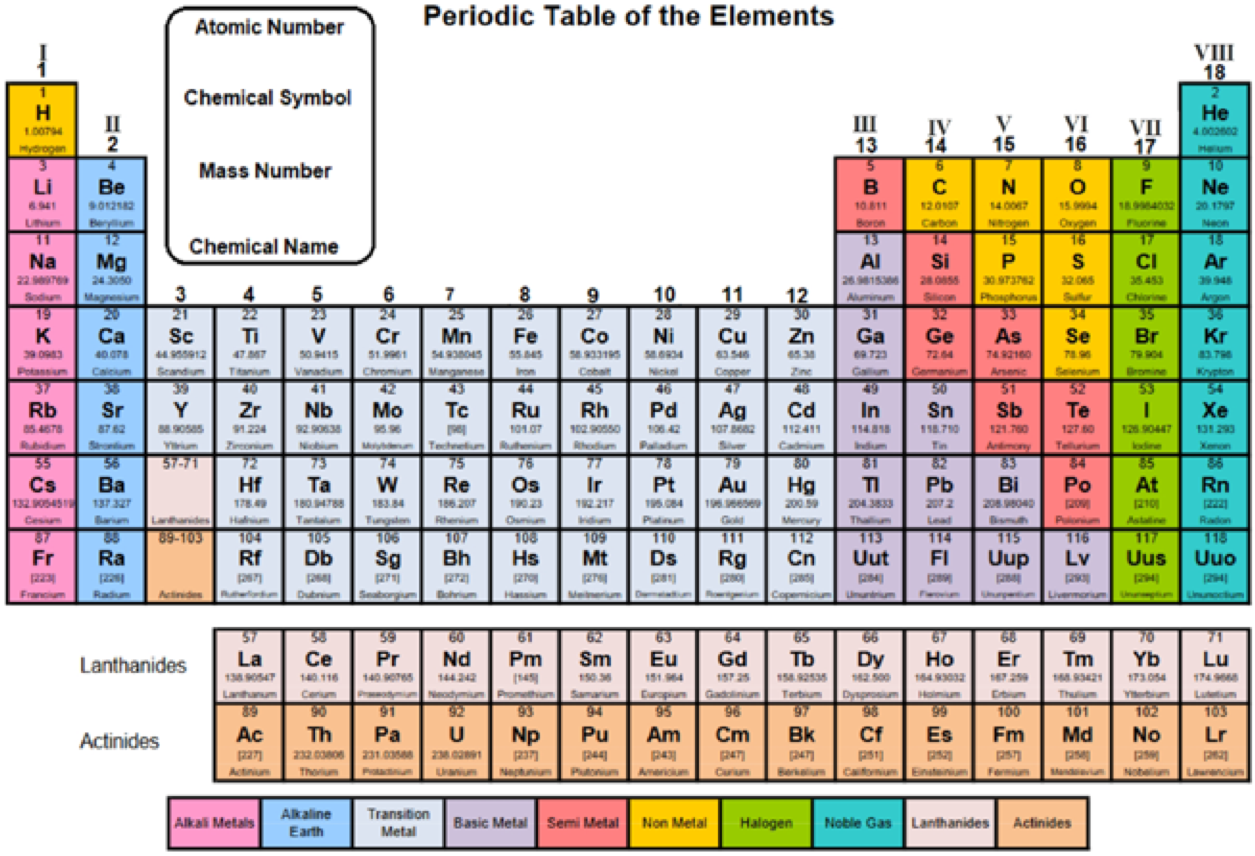


Figure 6.1: An example of one of the tiles on the Periodic Table

The diagram alongside shows an example of one of the tiles on the Periodic Table. Can you identify the element it represents? How many protons does an atom of this element have? What is the atomic mass number of the atom of this element?



There are different versions of the Periodic Table, which can each contain different information about the elements. Can you identify what information is provided about the elements in the following table?

The following Periodic Table only shows the symbols for the elements.

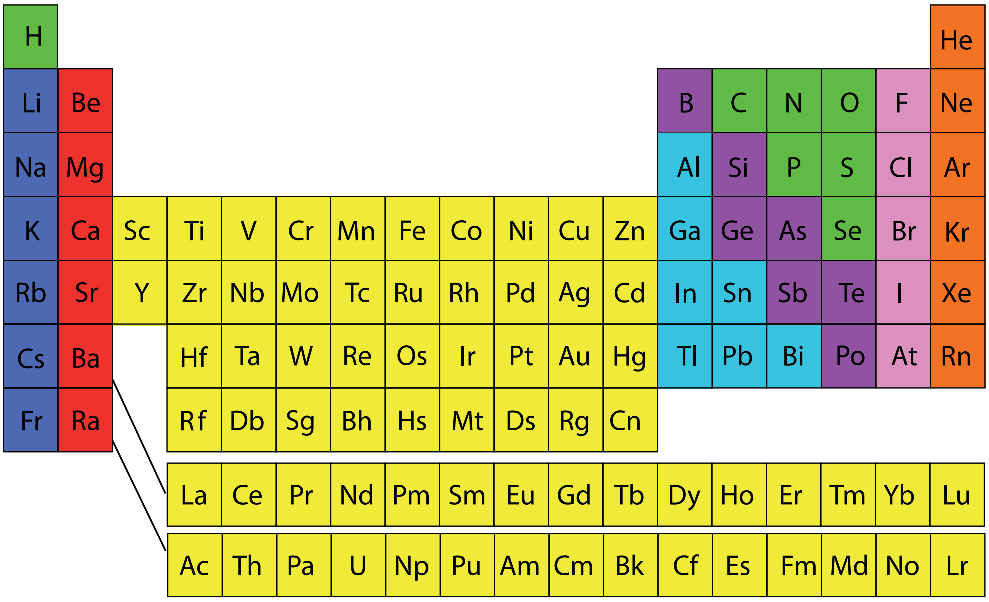


Figure 6.2: Periodic Table showing only the symbols for the elements.

Other versions of Periodic Table may contain additional information, such as:

- The element name; and/or
- The atomic mass, usually indicted at the bottom of each tile for an element.

The diagrams below show examples of how this information is sometimes presented.

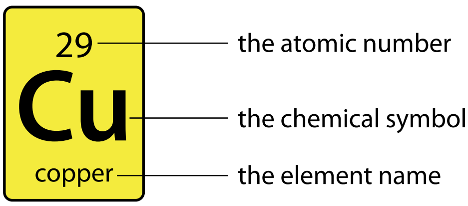


Figure 6.3: This tile shows information about the element copper.

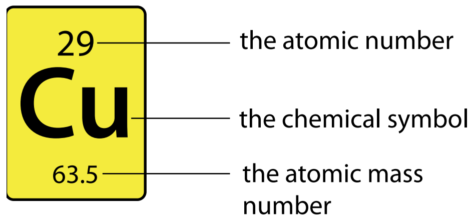


Figure 6.4: This tile also shows information about the element copper. Instead of the element name, the atomic mass of copper is given.

Keywords

- metal
- non-metal
- semi-metal
- group
- period
- electrons
- neutrons
- protons

How are the elements arranged in the Periodic Table?

We have learnt that the elements on the Periodic Table are arranged in a very specific way.

The elements are arranged in order of increasing atomic numbers. The element with the smallest atomic number is hydrogen (H: atomic number = 1) is in the top left-hand corner of the table. The elements with the largest atomic numbers are found at the bottom of the table.

The elements are also arranged in regions and these regions are often presented in different colours. The following Periodic Table shows us where the metals, non-metals and semi-metals can be found.

ACTIVITY [??]

Instructions

Look at the diagram below and answer the questions.

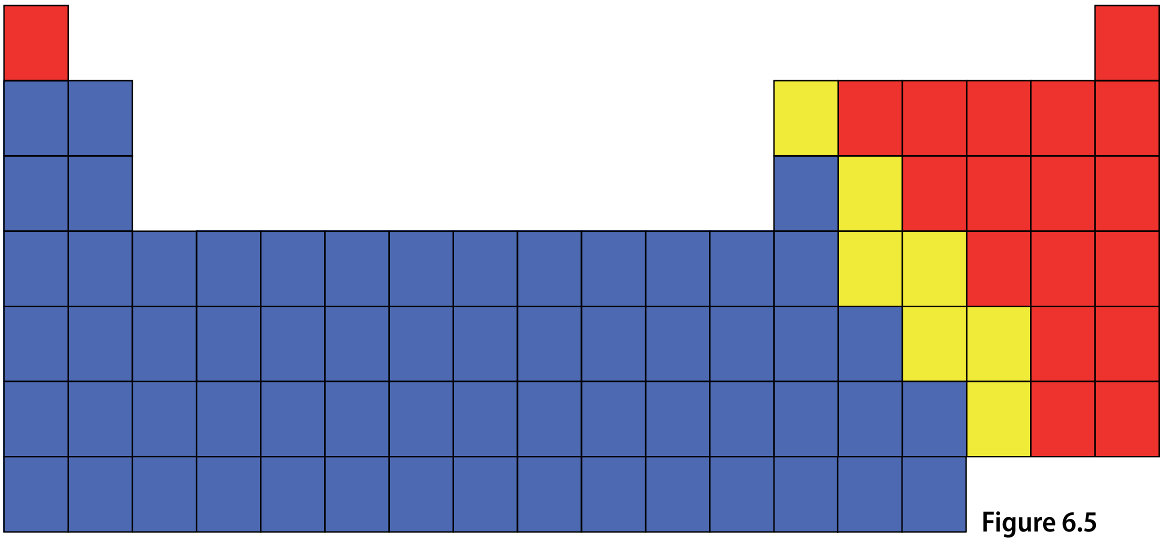


Figure 6.5

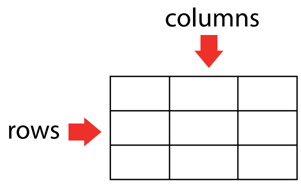
1. Where do we find metals in the Periodic Table?
2. Where do we find non-metals in the Periodic Table?
3. Where do we find semi-metals in the Periodic Table?

Take note

The semi-metals are also sometimes referred to as the metalloids.

All tables have rows and columns. Can you remember the difference between vertical and horizontal? Draw short lines to show the difference between 'vertical' and 'horizontal' in your exercise books.

Vertical runs 'up and down', and horizontal runs 'from side to side'. In a conventional table the columns run vertically, and the rows run horizontally.



There are special words to describe the columns and rows of the Periodic Table.

The following diagram shows what the column and rows are called.

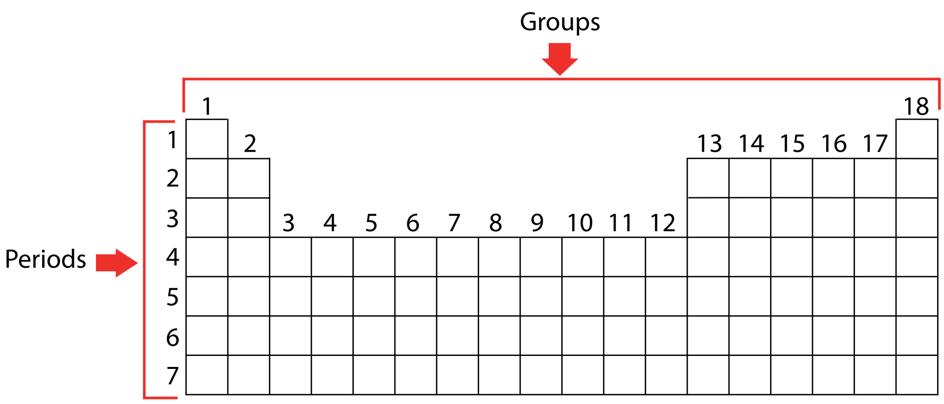


Figure 6.6: Diagram indicating Periods and Groups on the Periodic Table.

Groups: The vertical columns of the Periodic Table are called groups. The groups on the Periodic Table are numbered in such a way that Group 1 is on the left. The groups are numbered from 1 to 18. On older tables, the groups are numbered in a more complicated way. The colourful Periodic Table from Grade 7 (shown earlier) is an example of the numbering style that you may find in older textbooks and other science resources.

Periods: The horizontal rows of the Periodic Table are called periods. The first period is at the top of the table.

Names and chemical symbols

In Grade 7 we learnt that each element has a unique name. We also learnt that each element has a unique symbol. There is a list of simple rules to remember when using chemical symbols:

1. Every element has its own, unique symbol.
2. The first letter of the symbol is always a capital letter.
3. If the symbol has two letters, the second letter is always a small letter.
4. Some elements have symbols that come from their Latin names.

As scientists, we are expected to know the names and symbols of all the most important elements. You will not be expected to learn all of them off by heart just yet, but at the end of this unit you must know the names and chemical symbols of the first 20 elements on the table. To make them a little easier to remember, copy and complete the table below in your exercise books.

ACTIVITY Elements on the Periodic Table

Instructions

1. Use your Periodic Table to complete the following table.
2. Copy the table below in your exercise books. Write the chemical symbol and element name for each of the first 20 elements, identified by their atomic numbers.

Take note

You need to know the names and symbols of the first 20 elements on the Periodic Table, as well as iron, copper and zinc.

Atomic number	Chemical symbol	Element name
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		

Atomic number	Chemical symbol	Element name
13		
14		
15		
16		
17		
18		
19		
20		

3. There are three important industrial metals of which you need to know the names and formulae. Their atomic numbers have been written in the table below. Copy and complete the table below by filling in the chemical symbols and element names in your exercise books.

Atomic number	Chemical symbol	Element name
26		
29		
30		

Questions

1. What does the atomic number tell us about the atoms of an element?
2. How many protons are there in each oxygen atom?
3. How many neutrons are there?
4. In a neutral oxygen atom, how many electrons will there be?
5. What is the charge on protons and electrons?
6. How are the protons, neutrons and electrons (the sub-atomic particles) arranged in an atom?
7. Draw a model of an oxygen atom in your exercise books. Label your diagram.
8. What is the first element in the third period?
9. Which element is in Group 14 and in the second period? Write its symbol and its name.

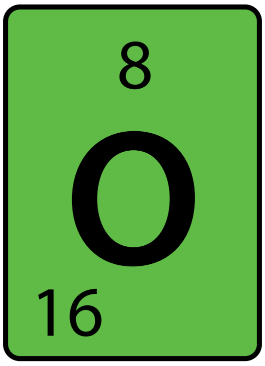


Figure 6.7

6.2 Elements and compounds

Can you remember learning about compounds in Grade 8 Matter and Materials?

We will start this unit by summarising and revising some of the main ideas about elements and compounds from Grades 7 and 8. This should help us to link the new ideas in this unit to what we already know.

Keywords

- compound
- crystal lattice
- diatomic molecule
- element

The particles that make up compounds

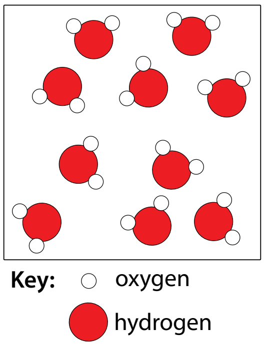


Figure 6.8: Water molecules.

The particles of a compound always consist of two or more atoms. In Physical Sciences Grade 10 you will learn that these atoms combine in different ways. In some cases they can form molecules. You may remember that 'molecule' is the word scientists use for a cluster of atoms that stick together in a specific way. Other compounds consist of atoms which are arranged in a regular pattern called a crystal lattice.

The molecules of a compound always consist of two or more different kinds of atoms, like the molecules of water in the diagram on the left.

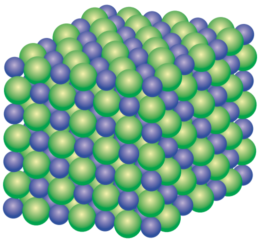


Figure 6.9: A sodium chloride crystal lattice consisting of sodium (purple) and chloride (green) atoms in a fixed ratio.

Compounds

Compounds that form crystal lattices consist of many atoms, but they always combine in a fixed ratio. For example, in sodium chloride (table salt), there is one chlorine atom for every sodium atom in the crystal. The smallest 'unit' that is repeated in the crystal consists of one Na and one Cl. The formula NaCl represents one 'formula unit' of NaCl.

From the diagram of the water molecules and the sodium chloride lattice on the left, we can see that a compound is not simply a mixture of elements. A mixture of the elements hydrogen and oxygen would look like the diagram on the right:

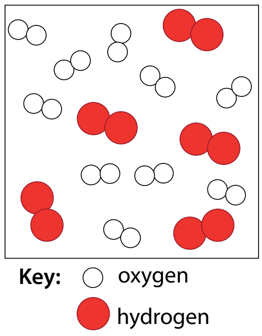


Figure 6.10: A mixture of hydrogen and oxygen molecules.

Why are the hydrogen and oxygen atoms paired in the diagram on the right? Before we answer that question, here is an important reminder: Elements are made up of just one kind of atom. Diatomic means 'consisting of two atoms'.

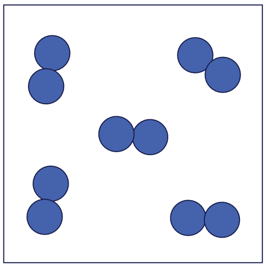


Figure 6.11: Some elements exist as diatomic molecules.

Some elements exist as diatomic molecules, like the ones in the diagram on the right below and the hydrogen and oxygen molecules in the 'mixture' diagram above. The most important examples of diatomic molecules are H_2, N_2, O_2, F_2, Cl_2, Br_2, and I_2.

Can you see that the water molecules in the diagram on the right are all identical?

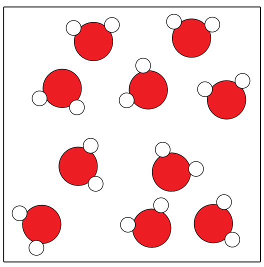


Figure 6.12: All water molecules consist of one O atom and two H atoms and this gives water its specific properties.

That brings us to the next point about compounds.

The atoms in molecules and lattices are combined in a fixed ratio.

In water, for example, one oxygen atom (O) has combined with two hydrogen atoms (H). All water molecules are exactly the same in this respect.

Any other combination of hydrogen and oxygen atoms would NOT be water. For example, hydrogen peroxide consists of the same elements as water (hydrogen and oxygen) but the ratio is different: two oxygen atoms have combined with two hydrogen atoms.

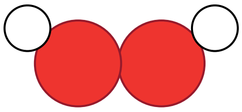


Figure 6.13: The hydrogen peroxide molecule consists of two O atoms and two H atoms. This gives hydrogen peroxide different properties from water.

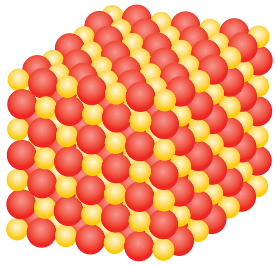


Figure 6.14: In the crystal lattice of black iron oxide, there is one iron (Fe) atom for every oxygen (O) atom.

The next important point about compounds is the following.

Each compound has a unique name and formula

Water can be represented by the formula H_2O.

The formula tells us that two hydrogen atoms (H) are combined with one oxygen atom (O) in a molecule of water.

What is the formula of hydrogen peroxide? Can you remember the name of the compound with the formula CO_2? Remember to take notes as you discuss things in class!

What formula represents one 'formula unit' of the type of iron oxide in the previous diagram?

Keywords

- chemical bond
- reactant
- product
- chemical formula

The atoms in a compound are held together by chemical bonds

What holds the clusters of atoms that we call molecules together? When atoms combine to form molecules, they do so because they experience an attractive force between them. The forces that hold atoms together are called **chemical bonds**.

Next, we need to be reminded where compounds come from.

Compounds form during chemical reactions

In all chemical reactions, the atoms in molecules rearrange themselves to form new molecules. This is how compounds form: the atoms in one set of compounds separate as bonds break between them, and they get rearranged into new groups as new bonds form. When this happens, we say a chemical reaction has occurred. Look at the following illustration.

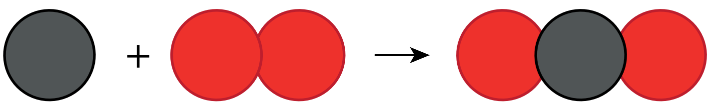


Figure 6.15

In the example above, the elements to the left of the arrow are called the **reactants**. They have rearranged to form a new compound. This is called the **product** and it is shown to the right of the arrow.

Can you describe what happened to the atoms and the bonds in this reaction?

Discuss which bond broke, which ones formed, and how the atoms were rearranged during the reaction.

The final aspect of compounds that we learnt about in Grade 8 is that each compound can be represented by a unique chemical formula.

A compound has a chemical formula

Compare the formula for water with the diagram of the water molecule you saw earlier. Can you make the connection?

The **chemical formula** of a compound is the same for all the molecules of that compound. When we read the formula, the subscripts tell us how many atoms of a particular element is in one molecule of that compound:

$$H_2O$$

The subscript '2' tells us that there are two H atoms in one molecule of water

O has no subscript; that means there is just one O atom in a molecule of water

Figure 6.16

When we write H_2O, we actually mean H_2O_1. According to convention, we do not use 1 as subscript in formulae, so the first formula is the correct one.

What it means is that there are 2 hydrogen atoms to every 1 oxygen atom. This is also a ratio and can be written as 2:1. We will practise writing formulae in the next activity.

ACTIVITY Writing formulae and revision

Instructions

1. In the following table, the names of some pure substances are given in the left-hand column. The middle column tells us what one molecule of each compound is made of.
2. You must use this information to write the formula of each compound in the final column, on the right.
3. The first row has been filled in for you, so that you have an example.
4. Column 1 contains the name: water.
5. Column 2: one molecule of water contains two H atoms and one O atom.
6. Column 3: From the information in column two we can write the formula: H_2O.

Name of substance	What it is made of?	Chemical formula
Water	2 H atoms and 1 O atom	H_2O
Carbon dioxide	1 C atom and 2 O atoms	
Ammonia	1 N atom and 3 H atoms	
Methane	1 C atom and 4 H atoms	

Questions

1. What holds the atoms together in a compound?
2. The following diagram shows how carbon and oxygen react to form carbon dioxide.

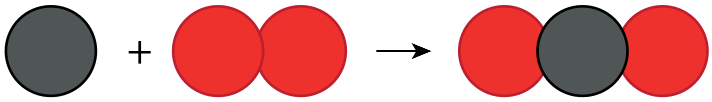


Figure 6.17

What are the reactants, and what is the product in this reaction? Copy the diagram into your exercise books and write these names onto it.

3. Why is oxygen represented as two circles together?
4. Magnesium oxide has the formula MgO. What does this ratio tell us about the atoms in the compound?

Now that we have refreshed our memories, we are going to return to the table that scientists use to organise their knowledge about the elements. Can you remember what it is called?

6.3 Names of compounds

Perhaps there are two or more people in your class with the same name? Then you will know how confusing it can be when two people have the same name! We have learnt that each element has a unique name. This is important, so that we do not end up confusing elements with each other.

Keywords

- systematic name
- suffix
- prefix

Each compound has a unique name

It is just as important for each compound to have a unique name.

When we write the chemical formulae for compounds, they are always a combination of the symbols of the elements in the compound. For example, when we see the formula NaCl, we know that this compound consists of Na and Cl.

When we name compounds, the names of the elements in the compound are combined and sometimes changed slightly, to make a name for the compound.

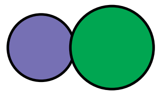
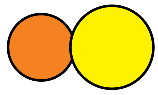
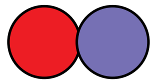
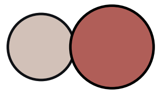
In the name sodium chloride, for instance, it is quite obvious that the compound being described must consist of sodium and chlorine. But, why is it chloride and not chlorine? Well, as you will see shortly, when we join up the names of the elements, the one that is named last is changed.

Take note

A 'suffix' is something placed at the end of a word. A 'prefix' is something placed at the beginning of a word.

Type 1: Compounds that contain a metal and a non-metal

For compounds of this type, the rule is simple. The metal comes first and the non-metal second. The name of the non-metal changes slightly: the suffix -ide replaces the ending of the name.

Formula	Consists of	Name	
NaCl	Sodium and chlorine	Sodium chloride	
FeS	Iron and sulfur	Iron sulfide	
MgO	Magnesium and oxygen	Magnesium oxide	
LiF	Lithium and fluorine	Lithium fluoride	

ACTIVITY Naming compounds of metals and non-metals

Instructions

1. Copy the table below in your exercise books and refer to the Periodic Table.
2. You need to identify the elements which make up the compound and give the name of the compound.

Formula	Which elements does it consist of?	Name
Li_2O		
KCl		
CuO		
NaBr		

Type 2: Compounds that contain only non-metals

This type of compound is slightly more complicated to name. There are three rules that you have to follow. They are as follows:

Rule 1:

The name of the element further to the left on the Periodic Table comes first, followed by the name of the element further to the right on the table. The name of the second element changes slightly: the suffix -ide replaces the ending of the name.

For example:

- oxygen becomes oxide
- fluorine becomes fluoride
- chlorine become chloride
- nitrogen becomes nitride.

Take note

The prefix mono- is usually left out from the beginning of the first word of the name. For instance, CO is carbon monoxide, not *mono*carbon monoxide.

Rule 2:

When two or more compounds have different numbers of the same elements (like CO and CO_2 in our example above), we must add prefixes to avoid confusion.

The first four prefixes are listed in the table below:

Number of atoms	Prefix
1	mono-
2	di-
3	tri-
4	tetra-
5	penta-

Here are some examples of how this rule should be applied:

Compounds of carbon and oxygen:
- CO – carbon monoxide (notice that it is not mono-oxide, but monoxide)
- CO_2 – carbon dioxide

Compounds of nitrogen and oxygen:
- NO_2 – nitrogen dioxide
- N_2O_4 – dinitrogen tetroxide (did you notice how tetra-oxide becomes tetroxide?)

Compounds of sulfur and oxygen:
- SO_2 – sulfur dioxide
- SO_3 – sulfur trioxide

We are going to practice what we have learnt so far in the next two short activities. First, we will write names from formulae.

ACTIVITY Writing names from the formulae of compounds

Materials

play dough, beans or beads

Instructions

1. How would you name the following compounds? Copy the table on the next page in your exercise books and write the name next to each formula.
2. Build one molecule of each compound with play dough, beans or beads. If you are not sure how to arrange the atoms, here is an important tip: the atom that comes first in the name (it will usually also be the first atom in the formula) must be placed at the centre of the molecule. All the other atoms must be placed around it. They will be bonded to the atom at the centre, but not to each other.

3. Draw a picture of your molecule in the final column of the table.

Formula of the compound	Name of the compound	Picture of one molecule of the compound
CO_2		
H_2O		
PF_3		
SF_4		
CCl_4		

Next, we will write formulae from the names of some compounds.

ACTIVITY Writing formulae from the names of compounds

Materials

play dough

Instructions

1. What formulae would you give the following compounds? Copy the table below in your exercise books and write the formula next to each name.
2. Build a model of each compound with play dough.
3. Draw a picture of one molecule of each compound in the final column of the table.

Formula of the compound	Name of the compound	Picture of one molecule of the compound
	hydrogen fluoride	
	dihydrogen sulfide	
	sulfur trioxide	
	carbon monoxide	

Take note

Now we have learnt an important new skill, namely to write and interpret the names and formulae of compounds.

There is one additional rule – an easy one to remember!

Rule 3:

Many compounds are not usually referred to by their systematic names. Instead, they have common names that are more widely known. For example, we use the name water for H_2O, ammonia for NH_3, and methane for CH_4.

In this unit we reviewed all the information about compounds and about the Periodic Table that we have learnt in previous years. We added some new information to both of these topics.

Summary

Key concepts

Elements

- All the atoms in an element are of the same kind. This means that an element cannot be changed into other elements by any physical or chemical process.
- Elements can be built up of individual atoms, or as bonded pairs of atoms called diatomic molecules.
- When atoms of different elements combine chemically, they form compounds.

Compounds

- In a compound, atoms of two or more different kinds are chemically bonded in a fixed ratio.
- The atoms that make up a molecule are held together by special attractions called chemical bonds.
- Compounds can be formed and broken down in chemical reactions.
- A chemical reaction in which a compound is broken down into simpler compounds and even elements is called a decomposition reaction.
- Compounds cannot be separated by physical processes, but they can be separated into their elements (or simpler compounds) by chemical processes.

The Periodic Table

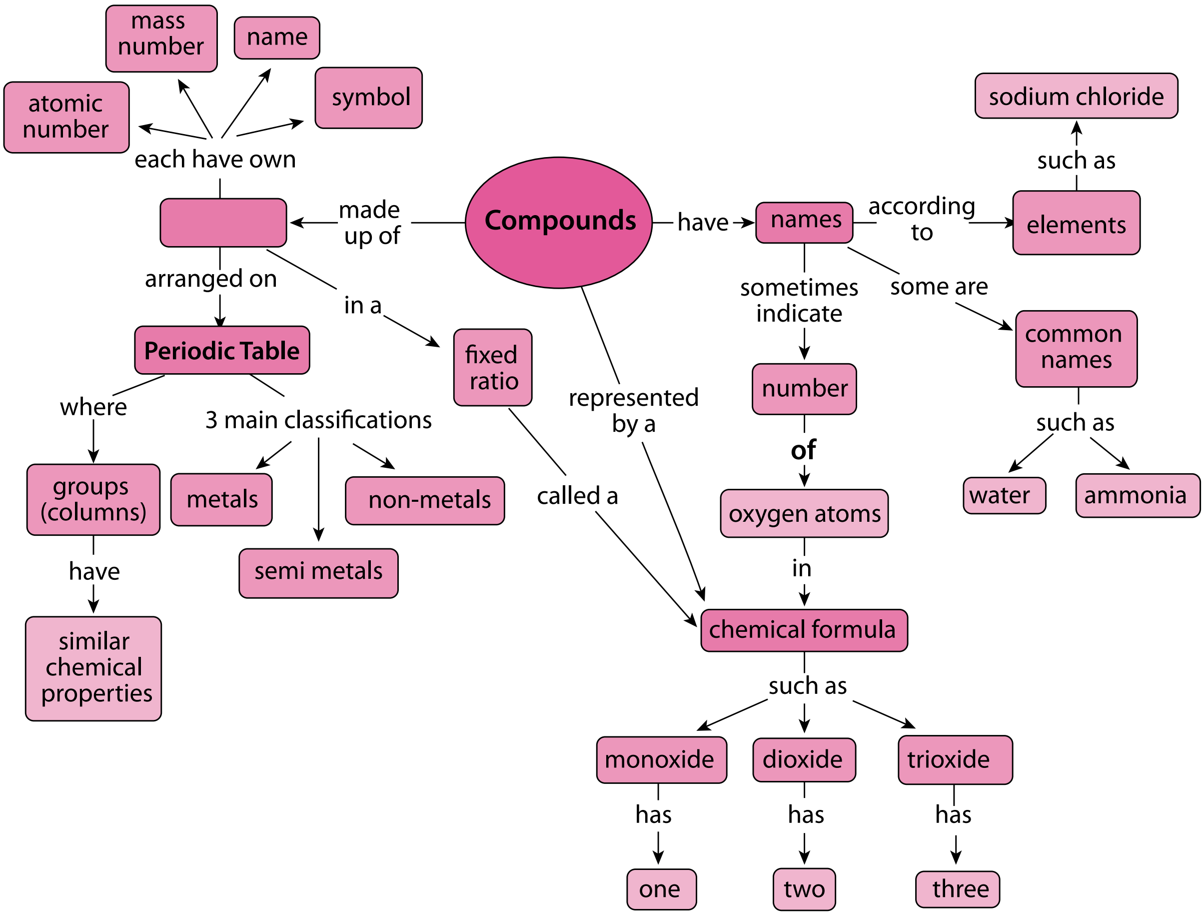
- Each element has a fixed position on the Periodic Table. The elements are arranged in order of increasing atomic number, with the lightest element (hydrogen: H) in the top left-hand corner.
- An element's position on the Periodic Table tells us whether it is a metal, a non-metal or a semi-metal:
  - metals are found on the left-hand side of the table;
  - non-metals are found on the far right-hand side of the table; and
  - semi-metals are found in the region between the metals and non-metals.
- An element can be identified in three different ways:
  - each element has a unique name;
  - each element has a unique chemical symbol; and
  - each element has a unique atomic number.
- The vertical columns of the Periodic Table are called groups. The Periodic Table has 18 groups.
- The horizontal rows of the Periodic Table are called periods. There are seven periods.
- Elements belonging to the same 'group' of the Periodic Table exhibit the same chemical behaviour, and will often have similar properties.
- Many different versions of the Periodic Table exist. Typically, the element symbol, the atomic number, and the atomic mass of each element are given on the table.

Names and formulae

- Each compound has a unique name and formula.
- The formula of a compound tells us which elements are in the compound and how many atoms of each element have combined to form one molecule of that compound.
- There are rules for naming compounds that take into account how many atoms of each type are in one molecule of the compound.

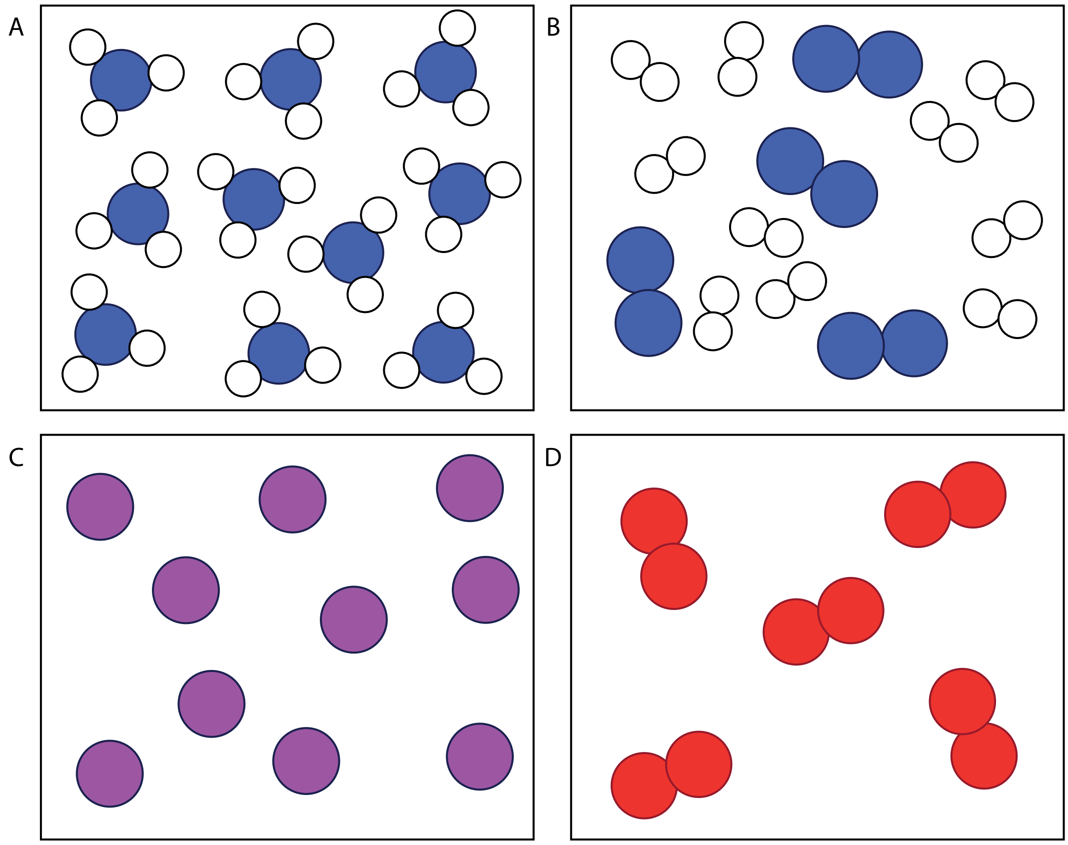
Concept map

Study the concept map below summarising what we have learnt in this unit about compounds.



Revision

1. Each of the four blocks below (labelled A to D) contain some matter. You must answer the following questions using the diagrams in the blocks. Each question may have more than one answer! [12]



a) Which block represents the particles of an element?
b) Which block represents the particles in a compound?
c) Which block represents the particles in a mixture?
d) Which block represents diatomic particles?
e) If the blue atoms are N and the white atoms are H, write the formula for the molecules in block A.
f) If the blue atoms are N and the white atoms are H, write the formula for the molecules in block B.
g) Which blocks contain molecules?
h) Which block contains single atoms?

2. How would you name the following compounds?
a) Copy the table below in your exercise books and write the name next to each formula.
b) Build a model of each compound with play dough.

c) Draw a picture of one molecule of each compound in the final column of the table. [12]

Formula of the compound	Name of the compound	Picture of one molecule of the compound
NH_3		
CO_2		
$CuCl_2$		
SO_2		

3. What are the formulae of the following compounds? [4]

Formula of the compound	Name of the compound
	sodium chloride
	dinitrogen monoxide
	sulfur trioxide
	carbon monoxide

4. Here is a balanced chemical equation. Answer the four questions below that relate to this equation: [8]

$CO + H_2 \rightarrow CO_2 + H_2$

a) Write the formulae of the reactants of this reaction.

b) Write the names of the reactants of this reaction.

c) Write the formulae of the products of this reaction.

d) Write the names of the products of this reaction.

5. The table below contains the chemical formulae of a few compounds. You have to write the number of atoms of each element(s) combined in one molecule of each compound. The first row has been filled in for you as an example. [8]

Chemical formula	What it is made of?
H_2O	2 hydrogen atoms and 1 oxygen atom
SF_4	
NO_2	
Fe_2O_3	
Na_2O	

Total [44 marks]

7 Chemical reactions

Key questions

- What is a chemical reaction?
- How can we represent what happens in a chemical reaction?
- What do the different symbols in a chemical reaction equation mean?
- What do the numbers in a chemical reaction mean?
- What does it mean to balance a chemical equation?
- How can we tell if a reaction is balanced?
- How do we translate between word equations, picture equations and chemical equations?

Keywords

- bond
- reactant
- product
- chemical reaction
- macroscopic
- submicroscopic
- symbolic

In Grade 8 Matter and Materials we learnt about chemical reactions for the first time. Can you remember the main ideas about chemical reactions? Here they are again:

During chemical reactions, materials are changed into new materials with new chemical and physical properties.

The materials we start with are called **reactants**, and the new materials that form are called **products**.

During a chemical reaction, atoms are rearranged. This requires that bonds are broken in the reactants and new bonds are formed in the products.

In this unit we are going to build on these ideas. We will focus on two things:

1. how to write chemical reaction equations; and
2. how to balance chemical reaction equations.

This will prepare us for the units that follow this one, in which we will be looking at different types of chemical reactions.

Before we get to chemical reactions, however, it is important that we remind ourselves of the different ways that we have been thinking about chemical compounds up to now. The next section will show how they all fit together

7.1 How do we represent chemical reactions?

Keywords

- chemical equation
- coefficient
- subscript

How would you define a chemical reaction? Write down some of your ideas.

The following words may help you formulate your sentences.

reactants, products, bonds, rearranged, atoms, molecules, new compounds

A chemical reaction is a rearrangement of atoms in which one or more compounds are changed into new compounds.

All chemical reactions can be represented by equations and models. To some people, chemical equations may seem very hard to understand. Since atoms and molecules cannot be seen, they have to be imagined, and that can be quite difficult! Luckily, we have had some preparation because we have been drawing molecules since Grade 7.

Any time that atoms separate from each other and recombine into different combinations of atoms, we say a chemical reaction has occurred. No atoms are lost or gained, they are simply rearranged.

Word equations

When we represent a chemical reaction in terms of words, we write a **word equation**. For example, when hydrogen gas reacts with oxygen gas to form water, we can write a word equation for the reaction as follows:

hydrogen + oxygen → water

To the left of the arrow, we have the 'before' situation. This side represents the substances we have before the reaction takes place. They are called the **reactants**. What are the reactants of this reaction?

The arrow shows that the chemical reaction has taken place.

To the right of the arrow we have the 'after' situation. This side represents the substances that we have after the reaction has taken place. They are called the products. What is the product of this reaction?

Picture equations

The same reaction of hydrogen reacting with oxygen can also be represented in pictures called submicroscopic diagrams. The diagram below shows that the atoms in two hydrogen molecules (H_2) and one oxygen molecule (O) on the left rearrange to form the two water molecules (H_2O) on the right of the arrow.

Hydrogen atoms are white circles and oxygen atoms are red circles.

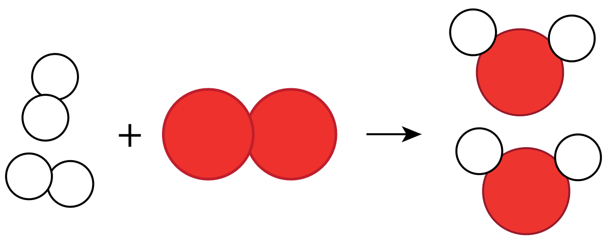


Figure 7.1

Questions

- 1. What is the product of the above reaction?
- 2. What are the reactants of the above reaction?
- 3. Write their formulae.

Chemical equations

When we represent a chemical reaction in terms of chemical formulae (symbols), it is called a **chemical equation**. The chemical equation for the above reaction would be as follows:

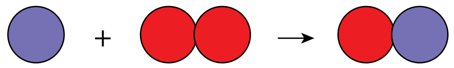
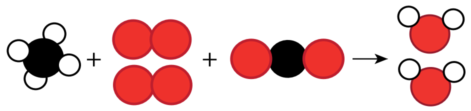
$$2H_2 + O_2 \rightarrow 2H_2O$$

On which side of the arrow are the reactants?

ACTIVITY Identifying the different types of equations

Instructions

Complete the following table by identifying the different types of equations which have been shown, namely word, picture or chemical equations.

Equation	Type of equation
	
carbon dioxide + water → glucose + oxygen	
$Fe + O_2 \rightarrow Fe_2O_3$	
	
$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$	

Coefficients and subscripts in chemical equations

When you look at the reaction equation above you will notice two kinds of numbers:

- Numbers *in front* of chemical formulae in the equation. They are called **coefficients**.
- Smaller numbers used inside and below the chemical formulae. These are called **subscripts**.
- Coefficients and subscripts mean different things, as you will see in the next section.
- Why is there a '2' in front of the formula for water (H_2O) in the chemical equation for water? This is because two molecules of H_2O can be made from two molecules of H_2 and one molecule of O_2 in our reaction.

Keywords

- balanced
- chemical formula
- rust
- tarnish

The numbers in front of the formulae in the chemical equation are called coefficients. They represent the numbers of individual molecules that are in the chemical reaction.

You will notice that O_2 does not have a coefficient in the reaction above. When there is no coefficient, it means that just one molecule of that substance takes part in the reaction.

In the previous unit, we learnt how to interpret chemical formulae. When we read the formula, the subscripts tell us how many atoms of a particular element are in one molecule of that compound.

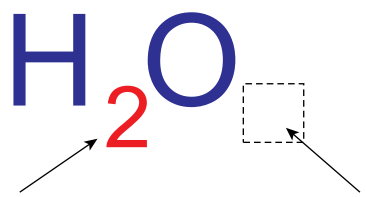


Figure 7.2

Visit

Simulation on balancing chemical equations.
bit.ly/17ImOVN

7.2 Balanced equations

Now we are going to learn what it means when a reaction is **balanced**. Here is our picture again.

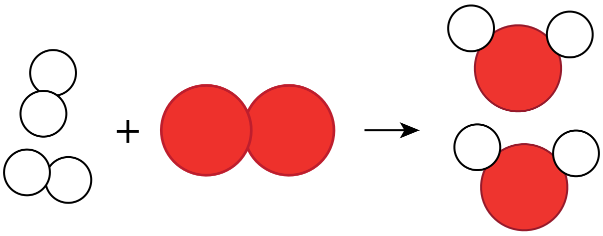


Figure 7.3

1. Count how many H atoms are on the left side of the reaction. How many on the right?
2. Count how many O atoms are on the left side of the reaction. How many on the right?
3. Did you notice that the numbers and types of atoms are the same on the left and on the right of the reaction?
4. The reactants have four H atoms and two O atoms.
5. The products have four H atoms and two O atoms.
6. When this is true of a reaction equation, we say the equation is balanced.

ACTIVITY When is a reaction balanced?

Instructions

1. Study the equation below. The black atoms are carbon (C), and the red atoms are oxygen (O). They will not always necessarily be this colour – this is just a representation.
2. Answer the questions that follow.

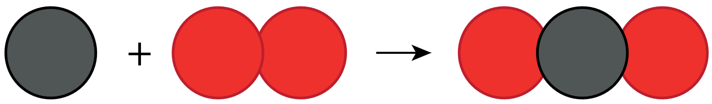


Figure 7.4

Questions

1. Write a symbolic representation (a chemical equation) for the above reaction.
2. Write the formulae for the reactants of this reaction.
3. Write the formula for the product of the reaction.
4. Count how many C atoms are on the left side of the reaction. How many on the right?
5. Count how many O atoms are on the left side of the reaction. How many on the right?
6. Is the reaction balanced? Why do you say so?

ACTIVITY Magnesium burning in oxygen

When magnesium metal burns in oxygen, we can write the following word equation for the reaction that occurs between these two elements:

magnesium + oxygen → magnesium oxide

Questions

Refer to the picture equation below and answer the questions that follow.

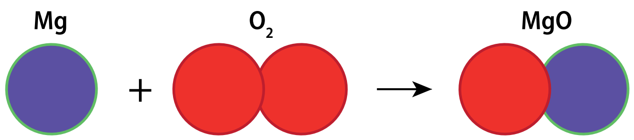


Figure 7.5

Number of atoms	Reactants	Products
Mg	1	1
O	2	1

1. From what you have in the table, is the equation balanced? Explain your answer.
2. Write a balanced the symbolic chemical equation.

Did you know?

Magnesium flakes are often used in some fireworks, such as sparklers, because when they burn they create bright, shimmering sparks.

Figure 7.6: Magnesium flakes burning in oxygen in a sparkler.

ACTIVITY Iron reacts with oxygen

When iron rusts, it is because the iron metal reacts with oxygen in the air to form iron oxide.

Figure 7.7: An old car with rust on the bonnet.

Figure 7.8: A close-up photo of a rusted barrel.

The word equation is the following:

iron + oxygen → iron oxide

The chemical equation is the following:

$$Fe + O_2 \rightarrow Fe_2O_3$$

Is the equation balanced? Draw a submicroscopic picture to help you decide.

You could also use a table like the one below:

Number of atoms	Reactants	Products
Fe		
O		

What is your verdict: Is the equation balanced? Explain your answer.

How could we balance the reaction? Three possibilities (Plans A, B and C) are given below. You must evaluate each plan, and say if it is allowed or not.

Plan A

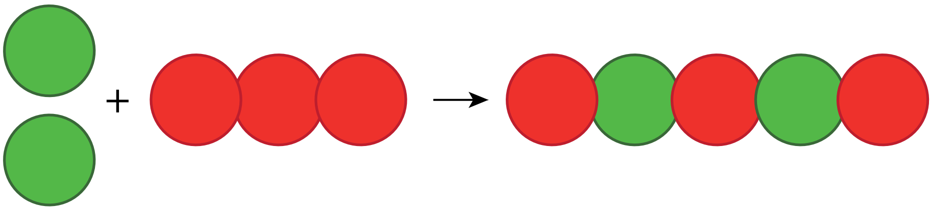


Figure 7.9

Changes made	Is this change allowed? Yes/no?	Reason
Add one Fe atom on the reactant side.		
Change O_2 to O_3 on the reactant side of the equation.		

1. Convert the picture equation above to a chemical equation.
2. Did any coefficients change? Remember that this is allowed.
3. Did any formulae change, or were any new formulae added? Remember that this is NOT allowed.
4. What do you think: Can this plan work? Explain your answer.

Plan B

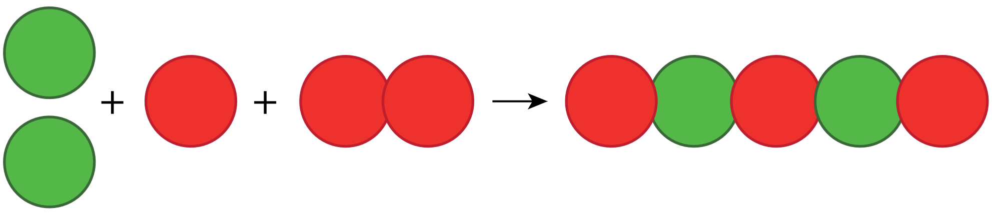


Figure 7.10

Changes made	Is this change allowed? Yes/no?	Reason
Add one Fe atom on the reactant side.		
Add one O atom on the reactant side.		

1. Convert the picture equation to a chemical equation.
2. Did any coefficients change? Remember that this is allowed.

3. Did any formulae change, or were any new formulae added? Remember that this is NOT allowed.
4. What do you think: Can this plan work? Explain why or why not.

Plan C

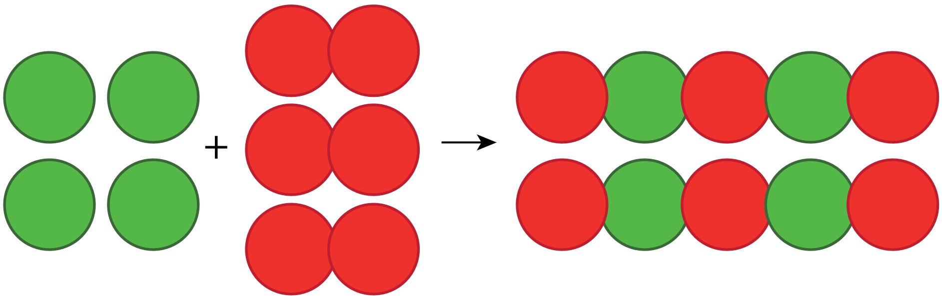


Figure 7.11

Changes made	Is this change allowed? Yes/no?	Reason
Add three Fe atoms on the reactant side.		
Add two O_2 molecules on the reactant side.		
Add one Fe_2O_3 on the product side.		

1. Convert the picture equation to a chemical equation.
2. Did any coefficients change? Remember that this is allowed.
3. Did any formulae change, or were any new formulae added? Remember that this is NOT allowed.
4. What do you think: Can this plan work? Explain why or why not.
5. Which of the three plans (A, B or C) helped us to balance the equation using only moves that are allowed?
6. Are there any other plans that you can think of to balance this equation?

In the next activity we will balance an equation that is much simpler, but we are not going to include all the explanations of the previous activity.

ACTIVITY Copper reacts with oxygen

Have you ever noticed how copper items tarnish over time?

This dark layer of tarnish is the result of a slow reaction between copper and oxygen to form copper oxide.

Figure 7.12: One of these copper coins is tarnished as it has become coated in a black substance.

Questions

1. Write the word equation for this reaction in your exercise books. The words are all in the sentence above; they just need to be placed in the correct positions.

	+		$\rightarrow$	

2. Convert the word equation into a chemical equation. You do not have to balance it yet.
3. Convert the chemical equation to a picture equation. It does not have to be balanced.

	+		$\rightarrow$	

4. Now, redraw the picture equation so that it is balanced. Remember that no 'new' compounds may be added; we are allowed to draw only more of the molecules that are already there.
5. Convert the balanced picture equation to a balanced chemical equation.

	+		$\rightarrow$	

In the units that follow, there will be more opportunities to write and balance chemical equations.

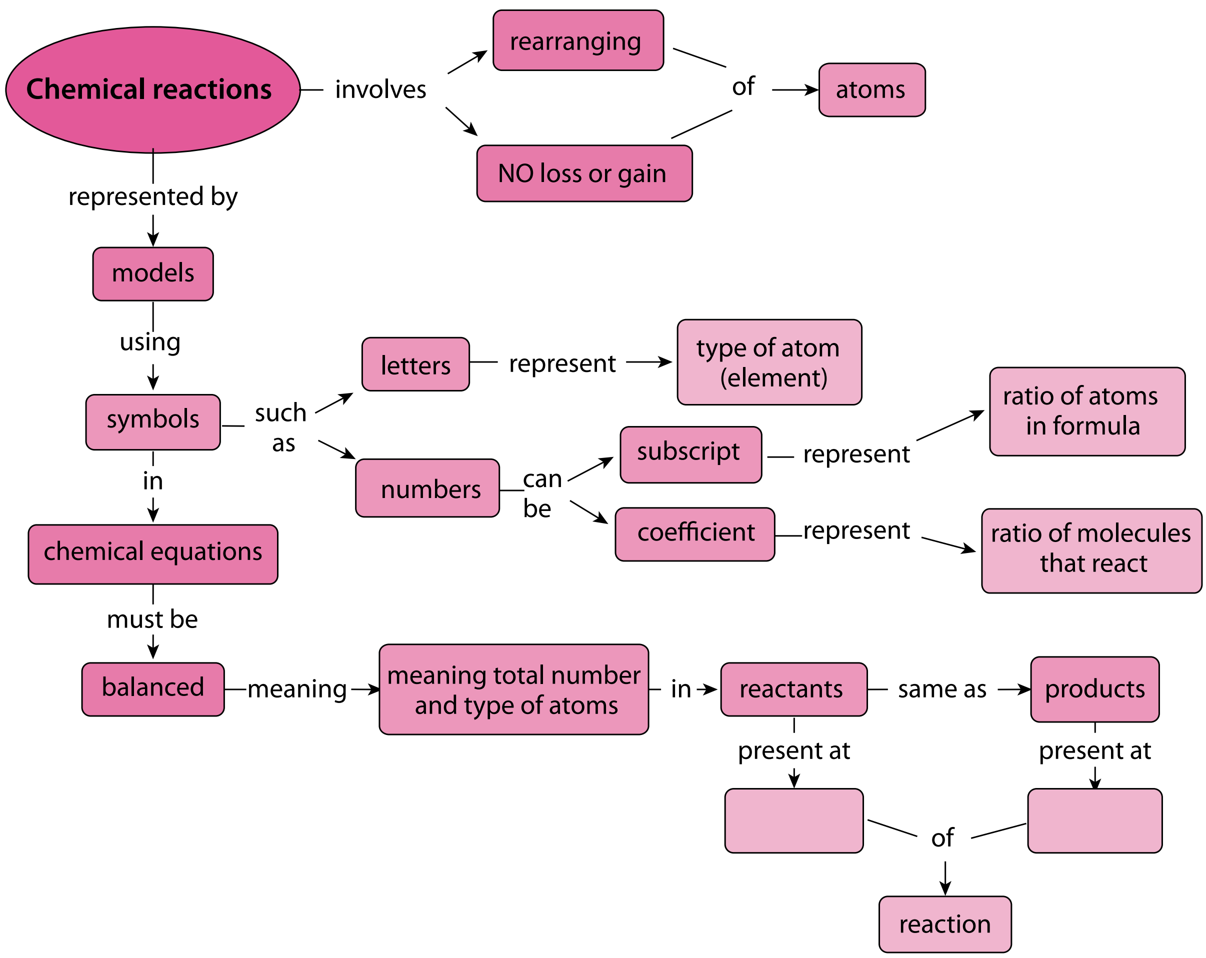
Summary

Key concepts

- There are a number of different ways to represent chemical equations:
  - With models and pictures (in submicroscopic representations);
  - with symbols and formulae (in chemical equations); and
  - with words (in word equations).
- Numbers are used in two different ways in chemical equations:
  - Coefficients in front of chemical formulae indicate the numbers of atoms or molecules of a specific type that take part in the reaction; and
  - Subscripts inside chemical formulae indicate the number of atoms of a specific type in that particular compound.
- Chemical reactions happen when atoms in compounds rearrange; no atoms are lost or gained during a chemical reaction.
- In a balanced equation equal numbers of the same kinds of atoms are on opposite sides of the reaction equation.

Concept map

The following concept map is incomplete. You need to describe when you get reactants and when you get products in a chemical reaction.



Revision

1. Why can we not change the subscripts in the formulae of reactants and products when we want to balance an equation? [2]

2. Write the balanced chemical equation between carbon and oxygen to form carbon dioxide. [1]

3. Write the balanced chemical equation between hydrogen and oxygen to form water. [1]

4. Here is a balanced chemical equation:
   $C + H_2O \rightarrow CO + H_2$
   Answer the four questions below that relate to this equation: [8]
   a) Write the formulae of the reactants of this reaction.
   b) Write the names of the reactants of this reaction.
   c) Write the formulae of the products of this reaction.
   d) Write the names of the products of this reaction.

5. The balanced equation below represents the reaction between nitrogen monoxide (NO) and bromine (Br_2):
   $2\ NO + Br_2 \rightarrow 2\ NOBr$
   Complete the table by counting how much of each atom is on each side of the reaction equation. [6]

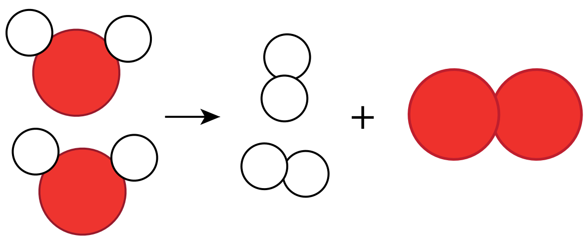
Number of atoms	In the reactants	In the product
Nitrogen (N)		
Oxygen (O)		
Bromine (Br)		

6. Turn the following chemical equations into word equations: [2 × 3 = 6]
   a) $2\ CO + O_2 \rightarrow 2\ CO_2$
   b) $2\ Mg + O_2 \rightarrow 2\ MgO$

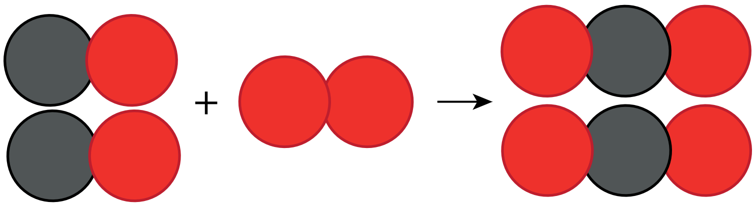
7. Turn the following word equations into chemical equations: [2 × 3 = 6]
   a) sulfur + oxygen → sulfur dioxide
   b) carbon monoxide + water → carbon dioxide + hydrogen

8. Turn the following picture equations into chemical equations. [2 × 3 = 6]
   - The red circles represent oxygen (O) atoms.
   - The white circles represent hydrogen (H) atoms.
   - The grey circles represent carbon (C) atoms.
   - The yellow circles represent sulfur (S) atoms

   a)



b)



9. Write the following chemical equations as word equations: [4 × 1 = 4]

a) $4\ Fe + 3\ O_2 \rightarrow 2\ Fe_2O_3$

b) What is the color of the product formed in a)?

c) $2\ Mg + O_2 \rightarrow 2\ MgO$

d) What is the color of the product formed in c)?

10. Turn the following chemical equations into picture equations: [2 × 4 = 8]

a) $CH_4 + 2\ O_2 \rightarrow CO_2 + 2\ H_2O$

b) $CS_2 + 3\ O_2 \rightarrow CO_2 + 2\ SO_2$

Total [48 marks]

8 Reactions of metals with oxygen

Key questions

- What happens when a metal reacts with oxygen?
- What is the product called?
- How can we represent the general reaction between a metal and oxygen?
- What is a combustion reaction?
- What is rust and how does it form?
- How can iron be made more rust-resistant?

Take note

The metals will react similarly with the other elements in the same group as oxygen (group 16).

In the previous unit, we learnt how to write and balance equations. The three examples we learnt about were:

- magnesium + oxygen → magnesium oxide
- iron + oxygen → iron oxide
- copper + oxygen → copper oxide

Which groups do magnesium, iron and copper come from?

In these reactions, the elements that react with oxygen are all **metals**. If you are not convinced of this, find them on the Periodic Table. Can you see that they are all found in the region occupied by the metals?

Metal Oxides

Where are metals located in the Periodic Table?

The names of the products of the three reactions above have something in common. Write down the names. Can you see what they have in common?

The products are all **metal oxides**. What exactly are metal oxides? As we will see later when we draw diagrams and write formulae to represent these reactions, they are compounds in which a metal is combined with oxygen, in a fixed ratio.

Keywords

- combustion

8.1 The reaction of iron with oxygen

We will be looking at how iron reacts with oxygen. In some cases, you may use steel wool for these experiments. Do you know what steel wool is? It is wire wool made of very fine steel threads. Steel is an alloy made mostly of iron. So, when we look at how steel wool burns in oxygen, we are actually looking at how iron reacts with oxygen.

Take note

A metal alloy is a solid mixture of two or more different metal elements. Examples are steel and brass.

Figure 8.1: Steel wool spinning creates interesting photos as the iron burns in oxygen and creates orange sparks.

When a substance burns in air, the reaction is called a **combustion** reaction. When a substance combusts in air, it is really reacting with oxygen.

ACTIVITY The reaction of iron with oxygen

Materials

- Bunsen burner or spirit burner
- matches
- safety goggles
- steel wool
- tongs

Instructions

1. Your teacher will demonstrate the combustion of iron in oxygen (which is present in air).
2. You should make careful observations during the demonstration and write these down in your exercise books. To guide you, some questions have been provided.

Questions

1. We used steel wool in this demonstration, but what is steel wool mostly made of?
2. Look at the metal before it is burned. Describe what it looks like.
3. Can you see the oxygen that the metal will react with? Can you describe it?
4. What do you observe during the reaction? Describe anything you see, hear, or smell.
5. What does the product of the reaction look like? Describe it in as much detail as possible.

If you think the reaction when iron burns in oxygen is spectacular, the next demonstration will amaze you!

Keywords

- camera flash
- ignite

8.2 The reaction of magnesium with oxygen

ACTIVITY The reaction of magnesium with oxygen

Materials

- Bunsen burner or spirit burner
- matches
- safety goggles
- magnesium ribbon
- tongs
- watch glass or beaker

Figure 8.2: Magnesium burns with a bright white flame.

Instructions

1. Your teacher will demonstrate the combustion of magnesium in oxygen.
2. You should make careful observations during the demonstration and write these down in your exercise books.

Questions

1. Describe the physical form (shape) of the metal in this experiment.
2. What do we call reactions where a substance burns in air?
3. How would you describe the physical appearance or colour of the metal before it is burned?
4. Can you see the oxygen that the metal will react with? Can you describe it?
5. What do you observe during the reaction? Describe anything you see, hear, or smell.
6. What does the product of the reaction look like? Describe it in as much detail as possible.

Magnesium is in group 2 in the Periodic Table. Do you remember that we said that elements in the same group will behave similarly? This means that they will react in a similar way. We have studied how magnesium reacts with oxygen, but calcium, for example, will behave in a similar way. You can watch the video in the visit link to confirm this.

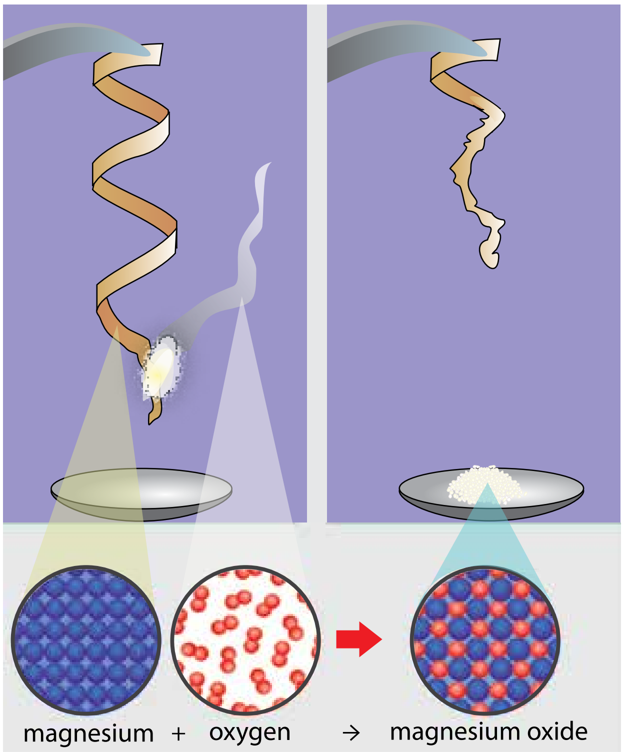


Figure 8.3: The diagram shows the combustion of magnesium to form magnesium oxide.

Did you know?

To take a photo in the dark, we need a camera flash. The earliest flashes worked with flash powder that contained magnesium grains. They had to be lit by hand and burned very brightly, for a very short period.

8.3 The general reaction of metals with oxygen

Let us start by writing word equations for the two reactions that we have just performed. Word equations are often easier to write than picture equations or chemical equations and so they are a good starting point when we want to write reactions.

Instruction

Write the word equation for the reaction between iron and oxygen and for the reaction between magnesium and oxygen.

The general equations

We can write a general word equation for reactions in which a metal reacts with oxygen:

metal + oxygen → metal oxide

When we use words to describe a reaction, we are still operating on the macroscopic level. Next, we are going to translate our word equation to a picture equation.

Keywords

- word equation
- picture equation
- chemical equation
- reactants
- product
- metal oxide

The chemical equation

We can show the reaction between magnesium and oxygen using a symbolic chemical equation.

$$2\,Mg + O_2 \rightarrow 2\,MgO$$

Since the chemical equation consists of symbols, we can think of this as a symbolic representation.

Did you know?

Group 1 metals are referred to as the Alkali Metals and Group 2 metals are referred to as the Alkaline Earth Metals.

Keywords

- rust
- corrosion
- corrosive
- rust-resistant
- steel

As we have said, the metals in the same group will react in the same way as each other with oxygen. So calcium reacts with oxygen in the same way as magnesium reacts with oxygen. The chemical equations also show similarities.

The chemical equation for the reaction between calcium and oxygen is:

$$2\ Ca + O_2 \rightarrow 2\ CaO$$

Questions

1. What is the product called in this reaction?
2. What group are calcium and magnesium from?

8.4 The formation of rust

Do you know what rust is? The pictures below will provide some clues.

Figure 8.4: Different objects which rust.

ACTIVITY The reaction between iron and oxygen in air

Materials

- test tube
- clamp
- retort stand
- dish
- iron filings
- water

Instructions

1. Rinse a test tube with water to wet the inside.
2. Carefully sprinkle a spatula of iron filings around the sides of the test tube.
3. Invert the test tube in a dish of water. Use a clamp attached to a retort stand to hold the test tube in place.
4. Over the three days the water must remain above the lip of the test tube.

On the right is a simple diagram showing the experimental setup with the clamp holding the test tube upright.

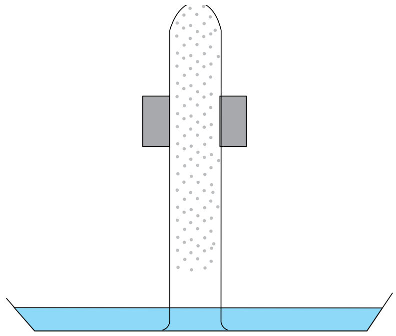


Figure 8.5: Experimental setup.

Questions

1. What do the iron filings look like at the start of the experiment?
2. What are the reactants in this experiment?
3. Is there something present that is aiding or speeding up the reaction?
4. What does the product look like at the end of the reaction?

Rust is a word to describe the flaky, crusty, reddish-brown product that forms on iron when it reacts with oxygen in the air. When your teacher burned the iron earlier, it reacted quickly with oxygen to form iron oxide. Here is a picture of iron oxide to remind you what it looked like.

Figure 8.6: A sample of iron oxide.

Rust is a form of iron oxide

When iron is exposed to oxygen in the air, a similar reaction occurs, but at much slower rate. The iron is gradually 'eaten away' as it reacts slowly with the oxygen. Under wet or moist conditions iron will rust more quickly.

Rust is actually a mixture of different oxides of iron, but the Fe_2O_3 of our earlier example is an important part of that. The rusting of iron is actually a good example of the process of **corrosion**. The equation for rusting is:

$$4Fe + 3O_2 \quad 2Fe_2O_3$$

Rusting tends to happen much faster near the ocean. Not only are there water droplets, but these droplets have salt in them and this makes them even more corrosive. Rusting also happens more quickly in the presence of acids. Inside laboratories, or factories where acids are used or stored, the air is also very corrosive. When the air in a specific area contains moisture mixed with acid or salt, we refer to the area as having a **corrosive climate**.

Figure 8.7: An abandoned car quickly rusts and corrodes near the sea.

Figure 8.8: A garden sculpture that was intended to rust to give it more texture.

If you live in a corrosive climate, for example near the ocean, it is often better to make the window frames and doors of your house from wood instead of iron and steel, because wood does not rust. Many people also use aluminium as this metal does not rust.

8.5 Ways to prevent rust

Rust forms on the surface of an iron or steel object when that surface comes into contact with oxygen. The oxygen molecules collide with the iron atoms on the surface of the object, and they react to form iron oxide. If we wanted to prevent that from happening, what would we have to do? There are 3 ways: painting, electroplating and galvanising.

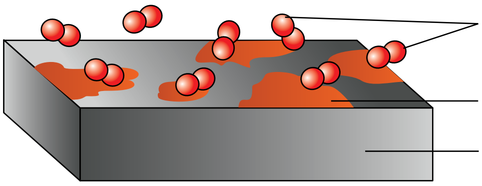


Figure 8.9: How rust forms

1. Paint provides a barrier to rust

If we wanted to prevent the iron atoms and oxygen molecules from making contact, we would need to place a barrier between them. That is what we are doing when we paint an iron surface to protect it from rust.

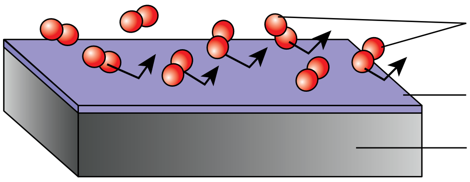


Figure 8.10: Painting to prevent rust

Paint is not the ultimate barrier, though. If the paint surface is scratched, or it starts to peel off, the metal will be exposed and rust can still form.

2. Electroplating

Electroplating is using electricity to coat with the metal with a metal that doesn't rust.

3. Galvanising

Galvanising is a process of applying a protective zinc coating to steel or iron to prevent rusting. The most common method is submerging the steel or iron parts in a bath of molten zinc.

The following diagram shows a segment of galvanised steel, with a scratch in the protective coating. What do you think will happen to the steel that is exposed to the air by the scratch in the coating?

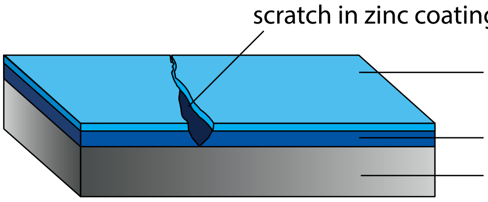


Figure 8.10: A segment of galvanised steel, showing damage to the zinc coating.

Keywords

- collide
- barrier
- exposed
- porous
- penetrate
- chromed metal
- galvanised metal
- galvanise
- oxidise

Did you know?

Zinc is in a different group from iron on the Periodic Table. This tells us that it does not react the same way as iron does with oxygen.

Iron that is galvanised is used for many different purposes. You will most probably have seen it being used as galvanised roof panels or other galvanised building materials, such as screws, nails, pipes, or floors.

Figure 8.11: Galvanised panels used for walls or roofs.

Figure 8.12: A galvanised watering can.

Did you know?

Cut apple slices turn brown as the iron compounds in the apple flesh are reacting with the oxygen in the air! The reaction is aided by an enzyme in the apple, so dripping lemon juice onto the slices will destroy the enzyme and prevent them from turning brown.

In this unit we learnt how metal oxides form. We saw two demonstrations of reactions in which metals oxides formed as products. Finally, we learnt about a metal oxide (iron oxide or rust) from our everyday experience as well as ways to prevent objects from rusting, especially those used in buildings and industry.

Figure 8.13: Galvanised nuts and bolts.

Figure 8.14 Galvanised flooring.

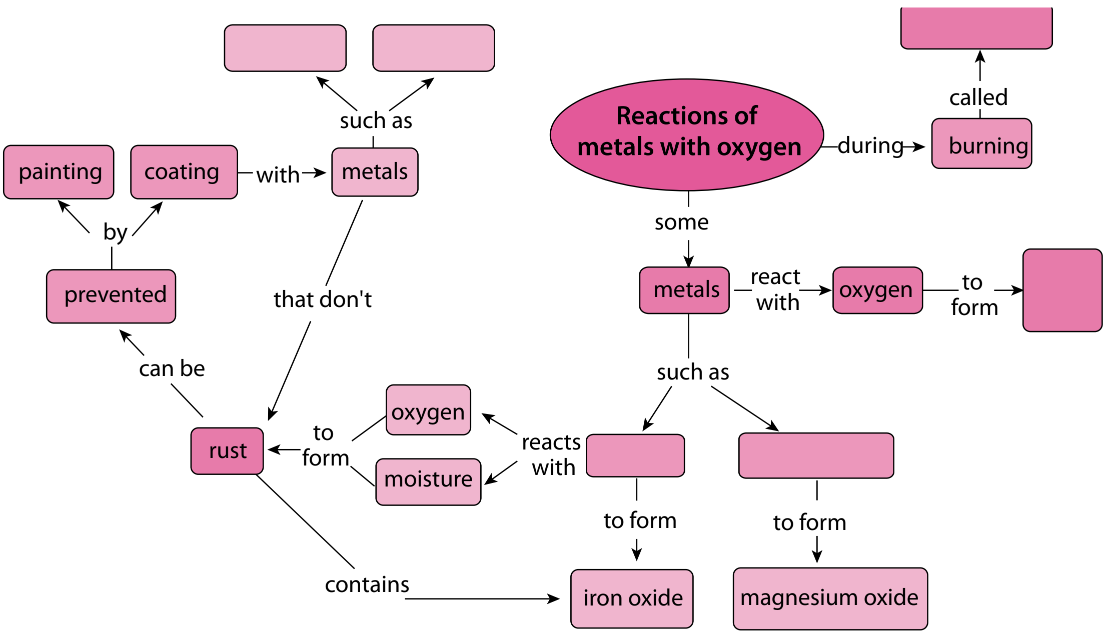
Summary

Key concepts

- When a metal reacts with oxygen, a metal oxide forms.
- The general equation for this reaction is: metal + oxygen → metal oxide.
- Some metals will react with oxygen when they burn. These reactions are called combustion reactions. Two examples of combustion reactions are:
  - Iron reacts with oxygen to form iron oxide:
    $4\ Fe + 3\ O_2 \rightarrow 2\ Fe_2O_3$
  - Magnesium reacts with oxygen to form magnesium oxide:
    $2\ Mg + O_2 \rightarrow 2\ MgO$
- Rust is a form of iron oxide, which forms slowly when iron is exposed to oxygen.
- Iron can be transformed to steel (an alloy), which is more resistant to rust.
- Rust can be prevented by coating iron surfaces with paint, or with rust-resistant metals such as chromium or zinc.

Concept map

What is the proper name for 'burning'? Fill this into the concept map. Fill in the examples of the metals that you studied in this unit. You will have to look at the products formed to know where to put which one. Lastly, give two examples of metals that you learnt about in this unit which do not rust.



Revision

1. Read the sentences and fill in the missing words. Write the completed sentences in your exercise books. [9]
   a) A chemical reaction where a compound and oxygen react during burning to form a new product is called a reaction.
   b) Magnesium + ____________ → magnesium oxide
   c) ____________ + oxygen → iron oxide
   d) copper + oxygen → ____________
   e) Another word for iron oxide is ____________
   f) Metal that is covered by a thin layers of zinc and zinc oxide is called ____________ metal.
   g) The gradual destruction of materials (usually metals) by chemical reaction with the environment is called ____________
   h) When the air in a specific area contains moisture mixed with acid or salt, we refer to the area as having a ____________ climate.
   i) The product of the reaction between a metal and oxygen is called a ____________

2. List three materials that can be used to protect iron or steel from corrosion. [3]

3. Copy and complete the table in your exercise books by providing the missing equations for the reaction between iron and oxygen. [4]

Word equation	
Chemical equation	
Balanced equation	

4. Copy and complete the table in your exercise books by providing the missing equations for the reaction between magnesium and oxygen. [4]

Word equation	magnesium + oxygen → magnesium oxide
Chemical equation	
Balanced equation	

5. Copy and complete the table in your exercise books by providing the missing equations for the reaction between copper and oxygen. [4]

Word equation	
Chemical equation	$Cu + O_2 \rightarrow CuO$
Balanced equation	

6. Copy and complete the table in your exercise books by providing the missing equations for the reaction between zinc and oxygen. [6]

Word equation	
Chemical equation	
Balanced equation	

Total [30 marks]

9 Reactions of non-metals with oxygen

Key questions

- What happens when a non-metal and oxygen react?
- How should we write equations for the reactions of carbon and sulfur with oxygen?
- Do all non-metals form dioxides with oxygen?

Oxygen is all around us in the air we breathe. It is a very **reactive** element. When an element is reactive, it means that it will readily react with many other substances. We saw evidence of the reactive nature of oxygen when we observed how it reacted with iron and magnesium to form metal oxides.

In this unit we look at a few reactions of non-metals with oxygen. Where do we find non-metals on the Periodic Table?

Keywords

- alternative
- fuel
- fossil fuel
- generate
- inert
- non-metal oxide
- non-renewable
- oxidise
- reactive
- renewable

9.1 The general reaction of non-metals with oxygen

When a non-metal burns in oxygen, a **non-metal oxide** forms as product. Here is the word equation for the general reaction:

non-metal + oxygen → non-metal oxide

Can you see that it looks similar to the word equation for the reaction between a metal and oxygen? The only difference is that the word 'metal' has been replaced with 'non-metal' on both sides of the equation. Non-metal oxides have different chemical properties from metal oxides. We will learn more about this later on in the term.

Take note

Substances that are not reactive are called unreactive or inert.

Let's look at a few specific examples of reactions in which non-metals react with oxygen. The first one is one that you are already familiar with, namely the reaction of carbon and oxygen.

9.2 The reaction of carbon with oxygen

Have you ever seen coal burning in air?

Coal is a form of carbon that is used as **fuel** for many different purposes. It is one of the primary **fossil fuels** that humans use to **generate** electricity for powering our industries, our activities and our living spaces. We will look at this in more detail next term in *Energy and Change*.

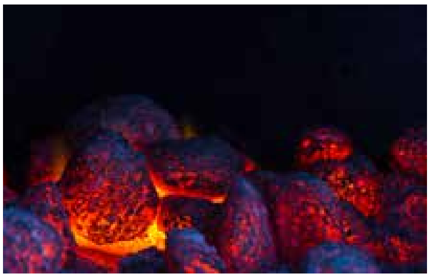


Figure 9.1: Coals burning in a fire.

Figure 9.2: A coal-powered power station.

ACTIVITY Coal burning in air

The energy in coal comes from the energy stored in plants and other organisms that lived hundreds of millions of years ago. Over the millennia, layers of dead plants and other biological waste were covered by layers of water and soil. The heat and pressure from the top layers caused the plant remains to turn into energy-rich coal.

Did you know?

37% of the electricity generated worldwide is produced from coal.

The energy released by burning coal is used to generate electrical energy in coal-powered power stations.

Coal is a form of carbon and when it burns in oxygen we can represent the reaction with the following word equation:

carbon + oxygen → carbon dioxide

Questions

1. Write a chemical equation for this reaction.
2. Balance the chemical equation you wrote.
3. In which group is carbon found in the Period Table of elements?

The other elements in the same group as carbon will react in the same way as carbon with oxygen.

Keywords

- toxic
- preservative

9.3 The reaction of sulfur with oxygen

What is the name of the product of the reaction between sulfur and oxygen? Use the name of the product and the general equation given at the start of the unit to complete the following word equation:

sulfur + oxygen →

Figure 9.3: Sulfur is a yellow substance, which burns with a blue flame in oxygen.

Figure 9.4: Sulfur mining is very dangerous to the miners who inhale the toxic sulfur dioxide gas.

Sulfur burns in oxygen to form sulfur dioxide. Your teacher will not demonstrate this reaction, because the sulfur dioxide that forms is a poisonous gas that you and your classmates should not be exposed to. Sulfur dioxide is sometimes used as a **preservative** for dried fruits, such as dried peaches and apricots and the guava rolls that many of us love to eat. The fact that it is toxic means that very small quantities of it can be added to food and drinks to preserve it. In very small quantities SO_2 does not permanently harm a large organism such as a human being, but bacteria cannot survive when it is present. Sulfur dioxide is also an important preservative in many South African wines.

Figure 9.5: Dried fruit, such as apricots, are preserved with. sulfur dioxide.

Figure 9.6: Many South African wines are preserved with sulfur dioxide.

ACTIVITY The reaction between sulfur and oxygen

In the following activity we are going to review word equations, picture equations and chemical equations, using the reaction between sulfur and oxygen as our context.

You wrote the word equation for the reaction between sulfur and oxygen above. Did you write the following?

sulfur + oxygen → sulfur dioxide

Questions

1. In what group do we find sulfur in the Periodic tTble of elements?
2. What are the reactants of this reaction? Write their names and formulae.
3. What is the product of the reaction? Write its name and formula.
4. Now, use the formulae of the reactants and products to write a chemical equation.
5. When is a reaction balanced?
6. Is your reaction above balanced? Why do you say so?
7. Draw a picture equation for the reaction, using the example of carbon above as guide.
8. Use play dough or clay to build models of the reactants and products of the reaction. This is what your starting reactants could look like:

Figure 9.7

Challenge question: How many bonds were broken and how many bonds were formed during this reaction?

Did you know?

The archaic (very old) name for sulfur is 'brimstone'.

Keywords

- dioxide
- systematic name

Take note

You may not cover this section in class as it is an extension.

9.4 Other non-metal oxides

We have looked at two examples of non-metals reacting with oxygen to form non-metal oxides. Both of our examples had a **dioxide** as product (carbon dioxide and sulfur dioxide). Do all non-metals form non-metal dioxides when they react with oxygen? What do you think?

Not all non-metal oxides are dioxides, as the following examples show.

The reaction between phosphorus and oxygen

Phosphorus is a very reactive non-metal. Can you remember what reactive means?

When phosphorus reacts with oxygen the chemical equation for the reaction is the following:

$$4\,P + 5\,O_2 \rightarrow 2\,P_2O_5$$

How many phosphorus atoms are in P_2O_5? How many oxygen atoms are in P_2O_5?

What is the **systematic name** of the product of this reaction? (If you are unsure how to name it, sneak a peek at Unit 6!)

Can you write a word equation for this reaction?

Our final example is a compound with which you should be very familiar!

The reaction between hydrogen and oxygen

Hydrogen and oxygen also react spectacularly. The reaction between a large quantity of hydrogen and oxygen in the air produces a beautiful orange fireball and a very loud boom! (You can watch the video in the visit box to see this in slow motion.)

Here is a diagram to show what is really happening to the compounds in this reaction. The purpose of the candle shown in the picture is to set the hydrogen gas alight, in other words, to provide enough energy for the reaction to start.

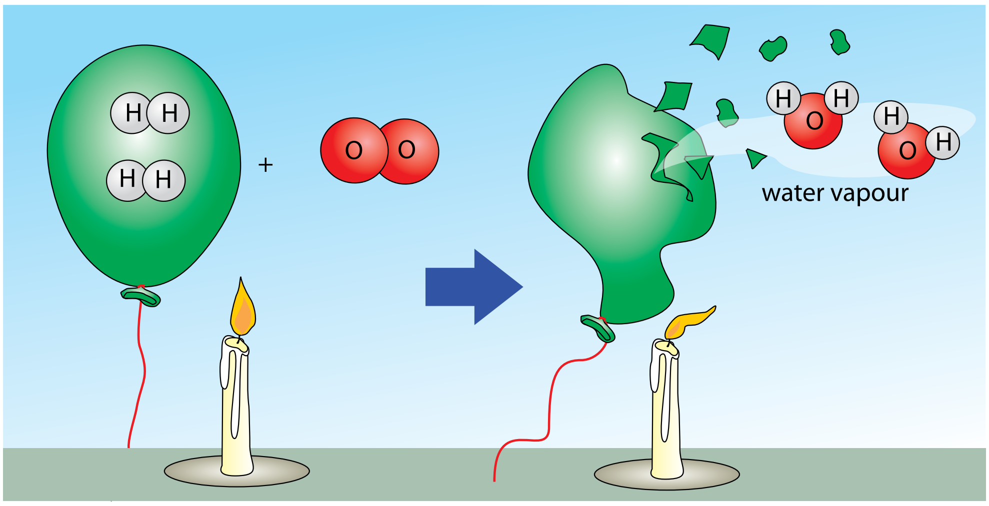


Figure 9.8

Can you complete the following chemical equation? The reaction is between hydrogen and oxygen. Write the product where it belongs.

$$2\ H_2 + O_2 \rightarrow$$

What is the common name of the product of this reaction? What is the systematic name of the product of this reaction? (If you are unsure how to name it, sneak a peek at Unit 6!)

Figure 9.9: In 1937, a German airship exploded and fell to the ground in a huge fireball as the hydrogen gas which kept it floating ignited and reacted with the oxygen in the air.

Did you know?

When a substance reacts with oxygen, we say it oxidises. It is called an oxidation reaction. Early chemists used the term oxidation for all reactions in which oxygen was a reactant. The definition of an oxidation reaction is broader and includes other reactions which you will learn about in Grade 10 if you continue with Physical Sciences.

Summary

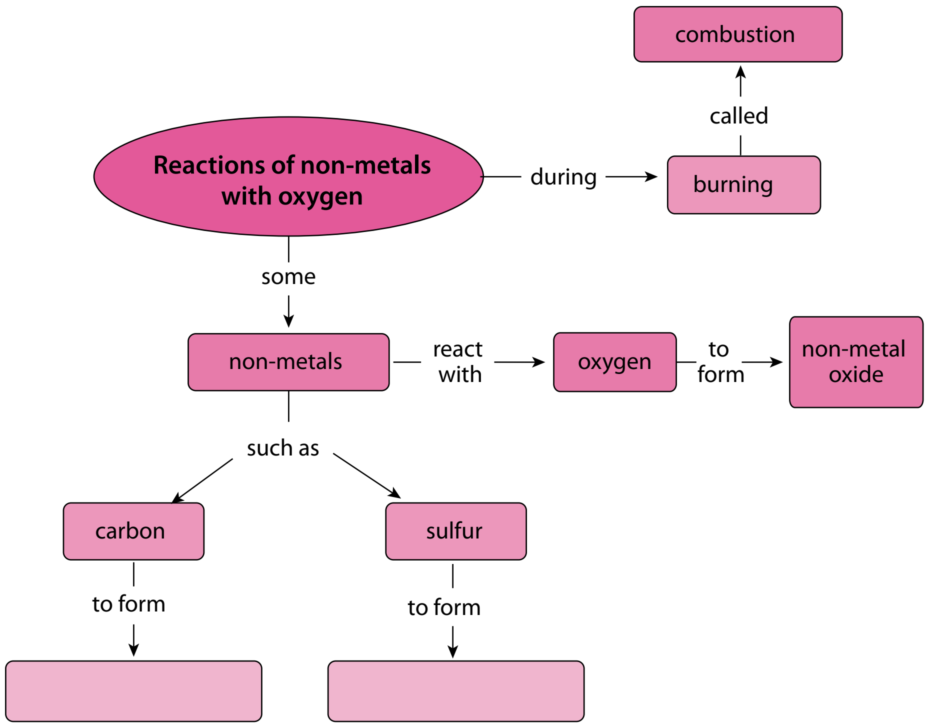
In this unit we have learnt about some of the reactions between non-metals and oxygen. Some of the skills that we practised during this unit were: writing equations (word, picture and chemical equations), and naming compounds.

Key concepts

- Non-metals react with oxygen to form non-metal oxides.
- The non-metal and oxygen gas (O_2) are the reactants in this type of reaction, and a non-metal oxide is the product.
- The reactions of carbon and sulfur with oxygen are examples of non-metals reacting with oxygen.
- Carbon and sulfur both form dioxides with oxygen, but this is not true of all non-metals.

Concept map

Complete the concept map on the following page. What will you fill in for the products when the two different non-metals react with oxygen during combustion?



Revision

1. Fill in the missing words in these sentences. Write the completed sentences in your exercise books. [5]
   a) A substance that will react readily with many other substances is called a ___________________ substance.
   b) Substances that do not react with other substances and do not change into other compounds are called ___________________ or ___________________.
   c) When a non-metal reacts with oxygen the product of the reaction is a ___________________.
   d) When a compound reacts with oxygen, we say it has become ___________________.

2. In your books, write a short paragraph (3 or more sentences) to explain what you understand each of the following terms to mean, in your own words. [3 × 3 = 9]
   a) systematic name
   b) preservative
   c) non-renewable energy source

3. For each of the following reactions, write a word equation, a chemical equation, and a picture equation in your exercise books.
   a) The reaction between carbon and oxygen. [6]
   b) The reaction between sulfur and oxygen. [6]

Total [26 marks]

10 Acids, bases and pH value

Key questions

- What measurement can we use to decide whether something is an acid or a base?
- What does 'the pH scale' refer to?
- How can we measure the pH of a substance?
- What does it mean if a substance has a pH below 7?
- What does it mean if a substance has a pH above 7?
- What does it mean when a substance has a pH equal to 7?
- How does a universal indicator respond to substances that are acidic, basic, or neutral?

Keywords

- unit
- acidity
- pH

10.1 What is the pH value?

In Grade 7 we learnt about acids and bases. Can you remember how to distinguish between them? Here is a table that highlights the main characteristics of acids and bases.

Acids	Bases
• Taste sour • Feel rough between your fingers • Can be corrosive • Can make bases lose their basic character • Turn blue litmus red	• Taste bitter • Feel slippery between your fingers • Can be corrosive • Can make acids lose their acidic character • Turn red litmus blue

We used the criteria in the table above to classify a number of substances as either acids, bases, or neutral substances. The table below contains some examples and shows their classification.

Acids	Bases	Neutral substances
Orange juice Vinegar Lemon juice Citric acid Gastric acid (stomach acid)	Bicarbonate of soda (baking soda) Soaps Bleach Ammonia solution	Distilled water Table salt solution Cooking oil

Finally, we learnt that there are substances that we can use that will show whether we have an acid or a base. Can you remember what they were called? Hint: They indicate, or show, whether we have an acid or base.

Indicators can show us if a substance is an acid or a base. In this unit we are going to link some important new learning to what we already know about acids and bases.

Measuring acidity and basicity

The numerical scale that we use to measure the **acidity** and **basicity** of a substance (how acidic that substance is) is called **pH**. We pronounce the two letters, 'p' and 'H' separately when we say pH.

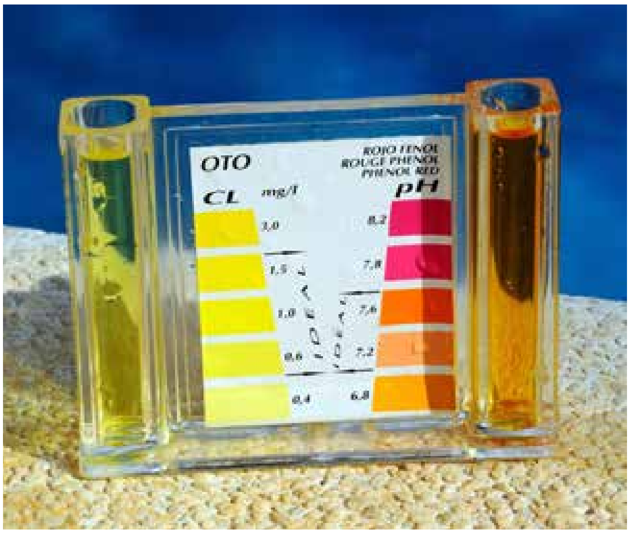


Figure 10.1: A kit for testing swimming pool water pH.

Have you ever heard the term pH?

Perhaps you have heard of a certain shampoo being 'pH balanced', or a skin soap that is 'neutral'. Perhaps you have heard that it is important for the water in a swimming pool to have 'the right pH'?

The pH scale ranges from the values of 0 and 14.

In science and in everyday life, we measure the acidity of substances in pH units. We could say that the 'acidity' of a specific shampoo has a pH of 5,5. pH is the unit of measurement and 5,5 is be the number indicating the relative acidity on the pH scale. It has become acceptable, however, for us to rather say: 'The pH of this shampoo is 5,5.'

In the next activity, we are going to get to know the pH scale a little better.

Did you know?

The term 'pH' was first described by Danish biochemist Søren Peter Lauritz Sørensen in 1909. The definite origin is disputed, but it is widely accepted that pH is an abbreviation for 'power of hydrogen' where 'p' is short for the German word 'potenz' (meaning power or exponent of) and H is the element symbol for hydrogen.

ACTIVITY The pH scale

Instructions

1. In the following picture the pH values of a variety of substances are shown on the pH scale.
2. Use the picture to answer the questions.

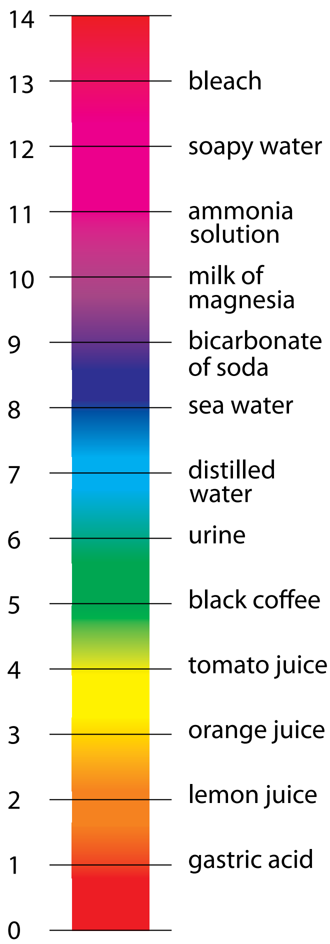


Figure 10.2: The pH scale.

Questions

1. Which of the substances in the table at the start of this unit can you find on the pH scale alongside? Copy the table into your exercise books and write their names and approximate pH values.

Name of substance	Approximate pH

2. Circle the names of all the acids in the table above with a red pen or koki.
3. Write the lowest and highest pH values of these acids. This represents the pH range of the acids on our list.
4. Does this range lie below or above pH 7?
5. Circle the names of all the bases in the table above with a blue pen or koki on the pH scale above.
6. Write the lowest and highest pH values of these bases below. This represents the pH range of the bases on our list.
7. Does this range lie below or above pH 7?
8. Find distilled water on the scale and circle it with a green pen or koki. Is water an acid or a base? Or is it perhaps something else?
9. What is the pH of water?
10. Which do you think is more acidic: orange juice or lemon juice? If you are not sure, ask yourself this question: Which one is more sour?
11. Which one has the lower pH: orange juice or lemon juice?

In the above activity we learnt a number of important things:

- Acids have pH values below 7;
- Bases have pH values above 7; and
- Neutral substances have pH values equal to 7.

This information has been summarised visually in the following diagram.

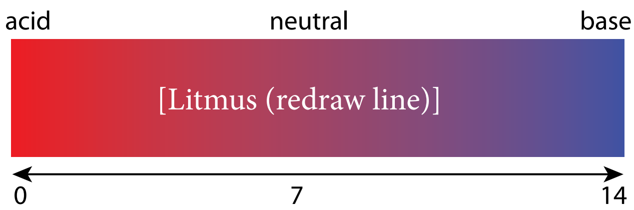


Figure 10.3

We saw in the activity that lemon juice, which is sourer than orange juice, has a lower pH than orange juice. Does that mean that the relative pH of a substance will tell us how acidic or basic it is?

Can we measure how acidic or basic something is?

When we compared orange juice and lemon juice earlier, we learnt something important: The lower the pH of a substance, the more acidic it is. For bases we can state the following: The higher the pH of a substance, the more basic it is.

Here is a summary:

- The closer to pH 1, the more strongly acidic the solution;
- The closer to pH 14, the more strongly basic the solution; and
- pH 7 is a neutral substance.

The pH value of a substance tells us if it is an acid or a base.

But how do we measure pH? One way to measure pH is with the help of acid-base indicators. Can you remember what they are? The next section will refresh your memory.

10.2 Indicators

Keywords

- indicator
- litmus
- universal indicator
- red cabbage indicator

What is an acid-base-indicator?

We know that some substances change colour when they react with an acid or a base. These substances are called acid-base indicators, which can show us if a substance is an acid or a base.

Different indicators change colour at different pH values. The table below shows a selection of acid-base indicators and the colours they will have at different pH values.

Indicator	Colour in acid (pH < 7)	Colour at pH = 7	Colour in base (pH > 7)
Red cabbage water	red, pink	purple	blue, green, yellow
Red onion water	red	violet	green
Tumeric water	yellow	yellow	red
Phenolphthalein	colourless	colourless	pink, red
Bromothymol blue	yellow	green	blue
Red litmus	red	red	blue
Blue litmus	red	blue	blue
Universal indicator	red, orange, yellow	green	blue, violet, purple

We made an indicator from red cabbage and even made some **red cabbage indicator** paper. Can you find red cabbage water on the table above?

- In acids, the red cabbage water will turn red or pink. In neutral solutions it will be purple or violet. Which colours will the red cabbage indicator be when it is mixed with a base?
- If red cabbage indicator is mixed with something that is only slightly basic, it will turn blue. If it is mixed with something that is strongly basic, it will turn yellow.

When you look at the table above and you compare the information given for red cabbage water with the picture on the right, the colour changes you observed in the red cabbage water will make sense!

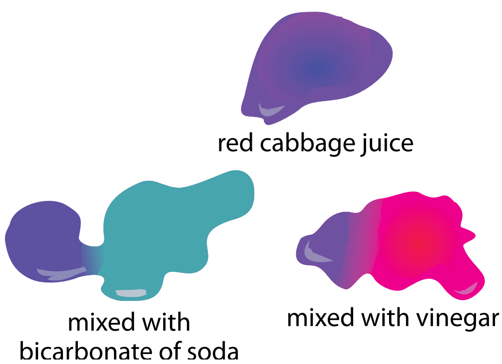


Figure 10.4: Red cabbage water mixed with base (left) and with acid (right). The blue drop at the top is the juice in a neutral solution (water).

You may recall that we also learnt about **litmus**, the most widely used of all acid-base indicators. Can you find litmus on the table of indicators?

Litmus does not change colour in the presence of a neutral substance, but responds to acids and bases in the following way:

- litmus is red in the presence of an acid; and
- litmus is blue in the presence of a base.

Litmus can be bought as a solution or as litmus paper, although the paper is more commonly used.

By changing to different colours in the presence of an acid or a base, indicators can show us if a substance is an acid or a base. In the next section we are going to learn about a special indicator that is so sensitive that it not only tells us whether a substance is an acid or a base, but also what its approximate pH is!

Figure 10.5: Blue and red litmus paper

Universal indicator

Unlike litmus, **universal indicator** can show us much more accurately how acidic or basic a solution is. Can you find universal indicator on the previous table of indicators? Universal indicator can change into a whole range of colours, depending on the pH of the solution. In the following picture, solutions of increasing pH were mixed with universal indicator to show its full range of colours.

Did you know?

Universal indicator can display so many colours because it is actually a mixture of several different indicators.

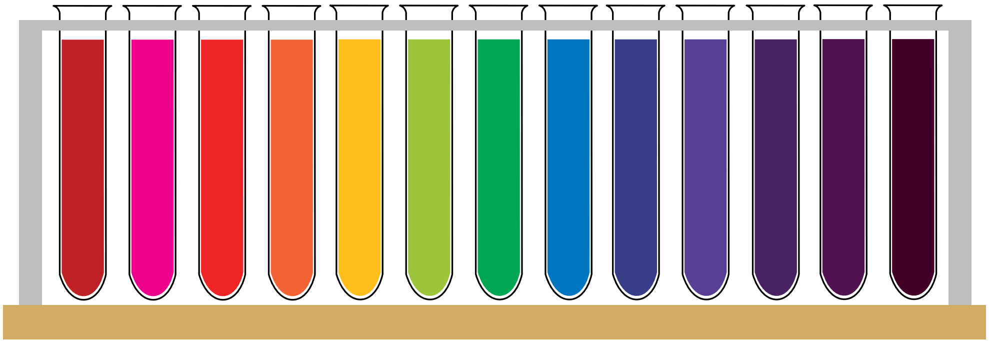


Figure 10.6: Universal indicator can have many different colours, from red for strong acids to dark purple for strong bases. The liquid inside the middle test tube is neutral (pH = 7), which is shown by the green colour of the indicator.

Like litmus, universal indicator also comes in paper form, with the pH colour range of the indicator printed on the packaging.

In the next investigation we will test a number of household substances with red cabbage indicator paper and with universal indicator paper.

Figure 10.7: Universal indicator paper

Investigation: Universal indicator paper and red cabbage indicator paper

The purpose of this investigation is to determine whether universal indicator and red cabbage can be used to show whether one substance is more acidic or basic than another.

Investigative question

What question are we trying to answer with this investigation?

Hypothesis

What do you think the answer to the investigative question is? You should try to make a prediction.

Identify variables

1. What will you be changing in this investigation? What is this variable called?
2. What will you be measuring in this investigation? What is this variable called?
3. What will you keep the same? What is this variable called?

Materials and apparatus

- small containers (test tubes or yoghurt tubs) containing the following substances:
  - distilled water
  - soda water
  - vinegar
  - lemon juice
  - sugar solution (1 tablespoon dissolved in a cup of water)
  - baking soda (bicarbonate of soda) (1 tablespoon dissolved in a cup of water)
  - Handy Andy (1 tablespoon dissolved in a cup of water)
  - aspirin (Disprin) (1 tablet in 2 tablespoons of water)
  - dishwashing liquid (1 teaspoon dissolved in a cup of water)
  - any other substances commonly used at home that are not dangerous.
- universal indicator paper
- red cabbage indicator paper
- glass or plastic rods (plastic teaspoons or straws will also work well)
- white tile or sheet of A4 printer paper.

Method

1. Use a small strip (1 cm long) of universal indicator paper for each substance that you will be testing. Place them on a sheet of printer paper or a white tile.

Take note

Do not use strong acids or bases, or bleach. Suggestions include: tea, coffee, rooibos tea, milk, tartaric acid, salt water, Sprite.

2. Dip the glass rod or straw into the first solution and transfer a drop of it to the first piece of universal indicator paper. Does the paper change colour? Write the colour of the paper with each substance in your table, in the appropriate place.
3. Compare the colour of the test strip with the colour range on the packaging of the universal indicator paper roll to find the pH of the solution. Write this in your table as well.
4. Rinse the straw very thoroughly with tap water before testing the next solution. Do so every time you move from one solution to the next.
5. Test all the solutions and record their colours.
6. Save the solutions to test them again with red cabbage indicator paper.
7. Use a small strip (2 cm long) of red cabbage paper for each substance that you will be testing.
8. Dip a fresh piece of paper into each of the test solutions and place it on the tile or white paper to dry. For each test solution, write the colour of the red cabbage paper in the table in the appropriate place.

Results and observations

Copy the table below into your exercise books and record your observations.

Substance	Colour with universal indicator paper	pH of the substance	Colour with red cabbage paper
Water			
Soda water			
Vinegar			
Lemon juice			
Sugar water			
Bicarbonate of soda			
Handy Andy			
Aspirin			
Dishwashing liquid			

1. Sequence the substances that you tested according to the colour change of the universal indicator, from the most acidic (darkest red) to the most basic (purple).

Questions

1. Which of the test substances are acids?
2. Which of the test substances are bases?
3. Which of the test substances are neutral substances?
4. Which substance is the strongest acid?
5. Which substance is the strongest base?

6. Count all the different colours that were possible with the red cabbage.
7. What colour(s) did the red cabbage paper turn in the test substances that were acids?
8. What colour(s) did the red cabbage paper turn in the test substances that were bases?
9. What colour(s) did the red cabbage paper turn in the test substances that were neutral?
10. Do you think red cabbage indicator can actually be used to measure pH? Why or why not?

Conclusions

What is your conclusion(s)? (Here you should answer the investigative question.)

Something to think about: Extension question

What could we do to make red cabbage indicator suitable for measuring pH?

In the last investigation we explored whether or not universal indicator paper or red cabbage indicator paper could tell us whether a substance is more acidic or basic than another. The advantage of using universal indicator over other indicators is that universal indicator can give us more accurate pH measurements. This is because it has different colours for different pH values. Most other indicators change colour only once or twice over the entire pH range.

Many other colourful foods can be used to make acid-base indicators. Check out the diagram below for some examples. You could even try out a few of them at home!

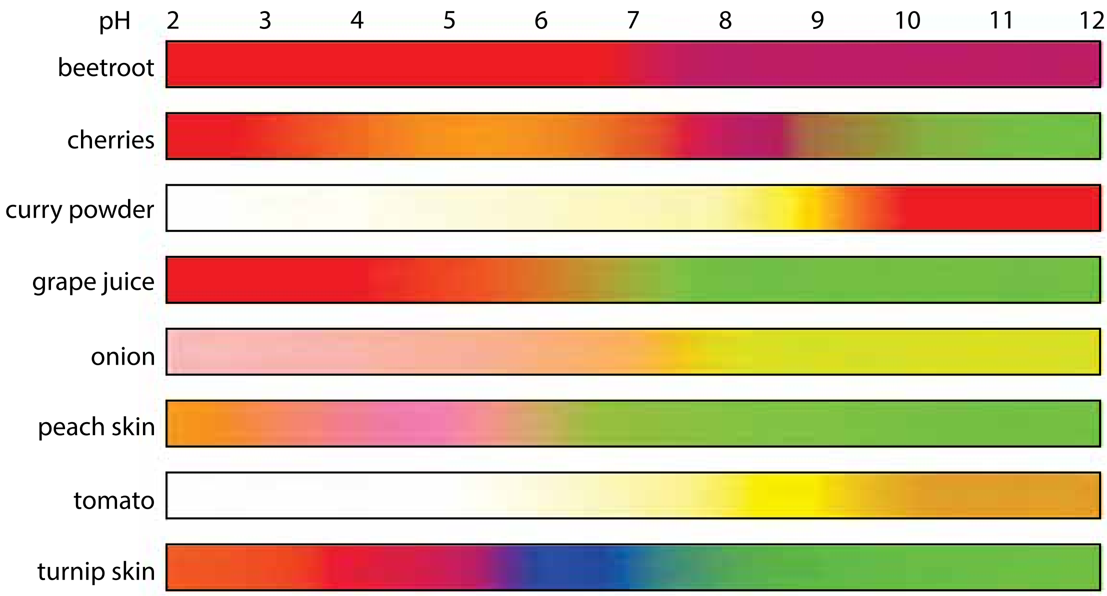


Figure 10.8: pH indicators made from edible substances.

Universal indicators give a range of colours that can be used to determine the pH of a solution. Litmus paper can indicate only whether a solution is acidic, neutral or basic.

Measuring pH with indicator solutions or paper is easy, economical and convenient if we have only a few measurements to make. If we have many pH measurements to make, tearing and dipping paper strips and matching them up with a colour chart can become quite tedious and time-consuming.

What other quick and easy ways are there to measure pH?

How else can we measure pH?

Scientists use a pH meter to quickly and accurately measure the pH of a substance. While they are much more expensive to purchase than indicator paper or solution, they are a worthwhile investment for a laboratory that has to make many pH measurements daily and needs these measurements to be done quickly.

Figure 10.9: A portable pH meter.

A pH meter is an electronic instrument with a special sensor at the end that is sensitive to acids and bases. This is more accurate than the universal indicator. Help the scientist to read the pH of the solutions in the photos and classify them as acidic, neutral or basic!

In this unit we have learnt about the pH scale. We have also learnt how to make pH measurements and how to interpret pH values.

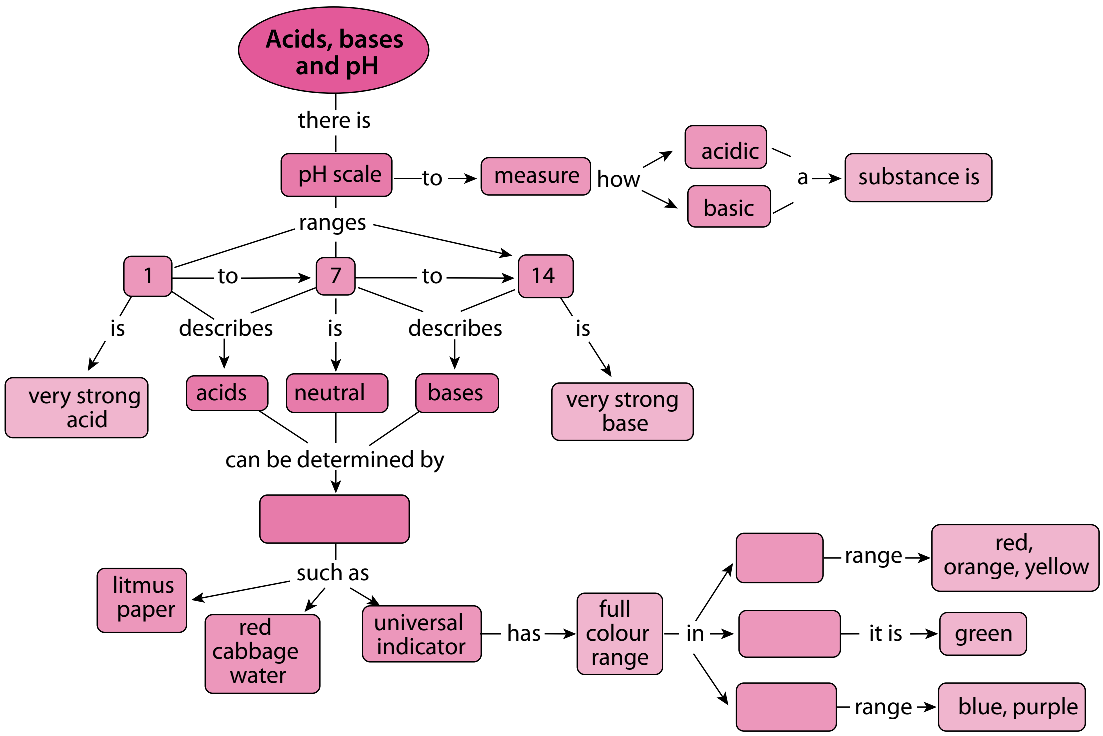
Summary

Key concepts

- When we want to decide whether a solution (in water) is acidic or basic, we can measure its pH.
- One of the ways pH can be measured is with an acid-base indicator, such as a universal indicator.
- An acid-base indicator is a substance that changes its colour depending on the pH of the solution to which it is added.
- The pH scale ranges between 0 and 14:
  - Acids have pH values lower than 7;
  - Bases have pH values higher than 7; and
  - Neutral substances have pH values approximately equal to 7.
- How acidic or basic a solution is, depends on its relative pH value:
  - The more acidic a solution is, the closer its pH value will be to 0; and
  - The more basic a solution is, the closer its pH value will be to 14.

Concept map

What can you use to determine whether a substance is an acid, base or neutral? Fill this in on the concept map. Finally, complete it by completing the information for the universal indicator. Fill in acid, base or neutral, depending on the colours listed.



Revision

1. Fill in the missing words in these sentences. Write the complete sentences in your exercise books. [6]

a) Something which shows whether a substance is an acid or a base, by changing colour when we add it to that substance, is called an ____________________.

b) The pH scale ranges between the values ____________________ and ____________________.

c) ____________________ have pH values less than 7.

d) Bases have pH values ranging between ____________________.

e) ____________________ substances have pH values approximately equal to 7.

2. Imagine we start with a beaker of clean, distilled water. Answer the following questions. [4]

a) What will be the pH of the clean, distilled water?

b) How will the pH change if we add a small amount of acid to the water?

c) How could we get the pH to increase?

d) How could we get the pH to increase to a higher value, for example 13?

3. In the following picture, the three beakers contain three different solutions. Red cabbage water was added to each of the beakers. Answer the following questions. [4 × 2 = 8]

A

B

C

a) Which solution, A, B or C, is the most acidic? Motivate your answer.

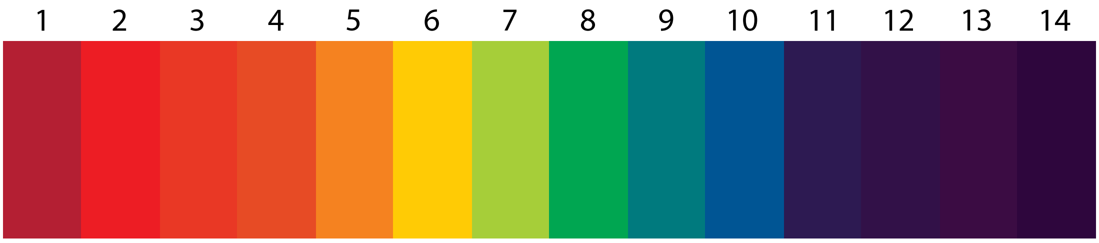
b) Which solution, A, B or C, is the most basic? Motivate your answer.

c) Which solution, A, B or C, is neutral? Motivate your answer.

d) What do you think would happen to the colour of solution A if we mixed it with solution B? Motivate your answer.

4. A scientist is given 6 solutions labelled A to F. The scientist tests each solution with universal indicator and records her results as follows:

Solution	Colour of universal indicator
A	Yellow
B	Blue
C	Green
D	Red
E	Purple
F	Orange



Use the results in the table and the colour guide for universal indicator underneath the table to answer the following questions: Copy the table below into your exercise books and answer the questions.

a) Which solutions are acidic? [2]

b) Which solutions are basic? [2]

c) Which solution is neutral? [2]

d) Copy the table below into your exercise books and use it to arrange the solutions in order from most acidic to most basic. Also write the colour and the approximate pH range of each solution. [6]

Solution	Colour of the solution	Approximate pH range of the solution

Total [30 marks]

11 Reactions of acids with bases

Key questions

- What is the reaction between an acid and a base called?
- What happens to the pH when an acid and a base are mixed?
- Does the reaction between an acid and a base always give a neutral mixture, in other words a mixture with pH = 7?
- Which factors will determine the pH of the final solution when an acid and a base are mixed?
- Is there a way to predict which classes of compounds will tend to be acids and which will tend to be bases?
- Are metal oxides, metal hydroxides and metal carbonates acidic or basic? Which pH range will their solutions fall into?
- What products can we expect when a metal oxide, a metal hydroxide or a metal carbonate react with an acid?
- Are there general equations to explain these reactions?
- How does acid rain form?

Keywords

- neutralisation reaction
- neutralise
- neutral solution
- potency
- detour
- exchange reaction
- laboratory acids
- corrosive
- acid rain

11.1 Neutralisation and pH

In the previous unit we learnt about a new concept, namely pH. If we want to know whether something is an acid or a base, we can measure its pH:

- Acids have pH values below 7. The lower the pH value, the more strongly acidic the substance.
- Bases have pH values above 7. The higher the pH value, the more strongly basic the substance.
- Neutral substances have pH equal to 7.

Another useful thing we learnt in the previous unit is that we can use universal indicator to measure the pH of a solution. Universal indicator has different colours at different pH values. Below is a colour chart showing the range of colours for universal indicator and the pH values they correspond to.

You will need it for all the activities of this unit, because we are going to do lots of pH measurements!

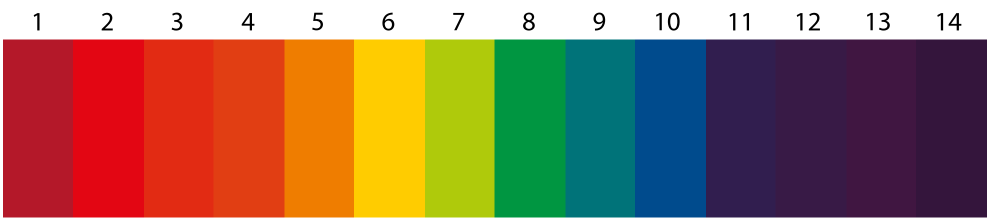


Figure 11.1

Can you remember how we used the universal indicator paper in the previous unit?

Is bicarbonate of soda an acid or a base? If you are not sure, turn back to the previous unit and look at the activity 'The pH scale'.

Investigation — The reaction between vinegar and bicarbonate of soda

The purpose of this experiment is to investigate how the pH changes when vinegar is added to bicarbonate of soda.

Investigative question(s)

What question do you hope to answer with this investigation?

Overview of the investigation

1. We will measure the pH of a solution of bicarbonate of soda with universal indicator paper to confirm whether it is acidic or basic. What range do you expect the pH of this solution to fall in?
2. We will add vinegar to the bicarbonate of soda solution in small portions and measure the pH after each portion has been added. What changes do you expect to observe? Will the pH increase, decrease or stay the same?

Hypothesis

What is your prediction? Your hypothesis should be a prediction of the finding(s) of the investigation. You should write it in the form of a possible answer to the investigative question.

Materials

- bicarbonate of soda
- vinegar
- water
- glass beaker or small yoghurt tub
- universal indicator paper (cut into 1 cm strips)
- sheet of white printer paper
- plastic teaspoon

Method

1. Prepare the universal indicator paper by neatly placing five 1 cm pieces underneath each other on the sheet of paper.
2. Place one teaspoon of bicarbonate of soda in the beaker or yoghurt tub.
3. Add approximately 10 teaspoons of water to the bicarbonate of soda.
4. Use the teaspoon to stir the solution until all the bicarbonate of soda has dissolved.
5. We will be calling this the 'test solution' from now on.
6. Transfer one drop of the test solution onto the first piece of universal indicator paper using the teaspoon or a straw.
7. Compare the colour of the paper with the colour guide given at the start of the unit, to find the pH of the solution. Record this pH in your results table.
8. Add 1 teaspoon of the vinegar to the test solution. Stir it gently and transfer another drop of the solution onto a fresh strip of the universal indicator.

9. Read the pH of the solution off the colour guide and record it in your results table.
10. Repeat steps 6 and 7 until the pH of the test solution drops below 7. You may need more than 5 pieces of universal indicator paper.

Results

Present your results in a neat table. Use appropriate headings for your table. 'Number of teaspoons of vinegar added' and 'Colour of the universal indicator paper' and 'pH of the test solution' are suggested headings for your columns.

Draw a line graph to illustrate your results. What will be on the x-axis and what will be on the y-axis? Give your graph a heading.

Conclusions

What conclusions can be made from the results of your investigation? Here you can rewrite your hypothesis, but change it to reflect your findings if they are different from what you predicted earlier.

Were you able to confirm or reject your hypothesis?

In this investigation, you probably noticed that the pH of the mixture dropped every time you added more vinegar to the bicarbonate of soda! Why did this happen?

When an acid and a base are mixed (in the right amounts), they will neutralise each other. That means that, together, they will change into something that is neither an acid nor a base. So, the acid will lose its 'acidity' and the base will lose its basicity.

What have we learnt so far? We have learnt that acids and bases neutralise each other:

- If we add a base to an acid, the pH of the resulting solution will increase, because the acid will lose some of its **potency**.
- If we add an acid to a base, the opposite will happen. The pH will decrease, because the base will lose some of its potency.

What are the products of an acid-base reaction? Can we predict what they will be?

The products of acid-base reactions

In order to understand how an acid-base reaction works, we have to take a quick detour and say something about exchange reactions. Acid-base reactions are **exchange reactions**.

In the reaction below, two substances AB and CD are undergoing an exchange reaction:

$$\mathbf{AB + CD \rightarrow AD + CB}$$

Can you see that A and C have exchanged partners so that A is now combined with D, while C combined with B?

What does this have to do with acids and bases? Well, acids and bases undergo exchange reactions too. Here are some examples. See if you can figure out which parts have exchanged with which.

Example 1

$$\mathbf{HCl + NaOH \rightarrow NaCl + HOH}$$

In the above equation HOH should actually be written: H_2O. The reaction equation becomes:

$$\mathbf{HCl + NaOH \rightarrow NaCl + H_2O}$$

or, in words:

hydrochloric acid + sodium hydroxide → sodium chloride + water

In this example, the following happened:
- the acid gave its H towards making a water molecule;
- the base gave OH towards making a water molecule; and
- the Na from the base and the Cl from the acid combined to form a salt.

Example 2

$$\mathbf{2\,HCl + MgO \rightarrow MgCl_2 + HOH}$$

In the above equation HOH should actually be written: H_2O. The reaction equation becomes:

$$\mathbf{2\,HCl + MgO \rightarrow MgCl_2 + H_2O}$$

or, in words:

hydrochloric acid + magnesium oxide → magnesium chloride + water

In this example, the following happened:
- the acid gave 2 H atoms towards making a water molecule;
- the base gave OH towards making a water molecule; and
- the Mg from the base and the 2 Cl atoms from the acid combined to form a salt.

Acid-base reactions always produce **water** and a **salt**. In both of the examples above the general equation was:

acid + base → salt + water

There is one class of acid-base reactions that produces an additional product, but we will learn more about that later.

Did you know?

In Grade 11 you will learn that the mechanisms of these reactions are actually slightly more complex than this, but for now, understanding it at this level is good enough.

Which laboratory acids should we know about?

When we investigated acids and bases in the previous unit, we considered only household acids such as lemon juice and vinegar. There are a few **laboratory acids** that you should know the names and formulae of. They are listed in the following table:

Name of the acid	Formula of the acid
hydrochloric acid	HCl
nitric acid	HNO_3
sulfuric acid	H_2SO_4

These acids are very **corrosive**, even when they have been diluted with water, and should always be handled with great care.

Figure 11.2: Hydrochloric acid is the acid we will be using in our investigations in this unit.

Figure 11.3: Look out for this label on bottles which contain corrosive substances, such as strong acids.

In the next sections will discuss the classes of substances that are typically acids or bases. Two important things to remember are the following:

- Non-metal oxides form acidic solutions when they are dissolved in water.
- Metal oxides, metal hydroxides and metal carbonates all form basic solutions when they are dissolved in water.

First, we will look at the non-metal oxides.

Did you know?

Carbonic acid is added to soft drinks to make them fizzy. The carbonic acid decomposes and forms carbon dioxide (CO_2)

Non-metal oxides form acidic solutions

Can you name a few non-metal oxides? Write down their formulae. If you are not sure you can take a peek at the Periodic Table and pick a few non-metals from the right-hand side of the table. Add oxygen and you have a non-metal oxide!

How do we know that non-metal oxides form acidic solutions? Experiments have shown this.

You may not know this, but when CO_2 gas is bubbled through water some of it dissolves in the water to form carbonic acid. Here is the reaction equation:

$$CO_2 + H_2O \rightarrow H_2CO_3$$

To see this happen, try the following quick activity.

ACTIVITY CO_2 bubbled through water

Materials

- tap water
- glass
- straw
- indicator paper

Instructions

1. Test the pH of clean tap water. It should be approximately 7. How would you do that?
2. Now exhale into the water using a straw. Your breath contains CO_2 and some of this will dissolve in the water if you carry on doing this for a few minutes.
3. If you measure the pH of the solution now, you will see that it has decreased! What do you think the pH will be?

Figure 11.5

The pH of the solution is now below 7 because it contains carbonic acid (H_2CO_3). Carbonic acid is not a very strong acid, but still acidic enough to have a pH lower than 7.

When sulfur dioxide (a gas) is bubbled through water it dissolves in the water to form an acid called sulfurous acid:

$$SO_2 + H_2O \rightarrow H_2SO_3$$

These are two of the reactions that produce a phenomenon called acid rain. SO_2 and CO_2 are released as waste products from factories and power stations. For example, burning wood and fossil fuels releases carbon dioxide and sulfur dioxide into the atmosphere. These gases then dissolve in water droplets in the atmosphere to form acids, in a similar way that the CO_2 in your breath dissolved in the water in the last activity to produce an acidic solution. When it rains, these acids are present in the raindrops that fall back to earth. Sulfurous acid (H_2SO_3) is strong enough to damage plant life and to acidify water sources.

Did you know?

Volcanoes also release non-metal oxides into the air (mainly SO_2) that can contribute to acid rain.

Figure 11.6: Acid rain forms when CO_2 and SO_2 from factories and other air pollutants combine with water in the atmosphere.

Figure 11.7: A forest that has been destroyed by acid rain.

ACTIVITY What is acid rain?

For the next activity, you have to do some research on acid rain.

Instructions

1. Study the diagram showing how acid rain forms.
2. Do some extra reading and research about acid rain.
3. Answer the questions about acid rain.

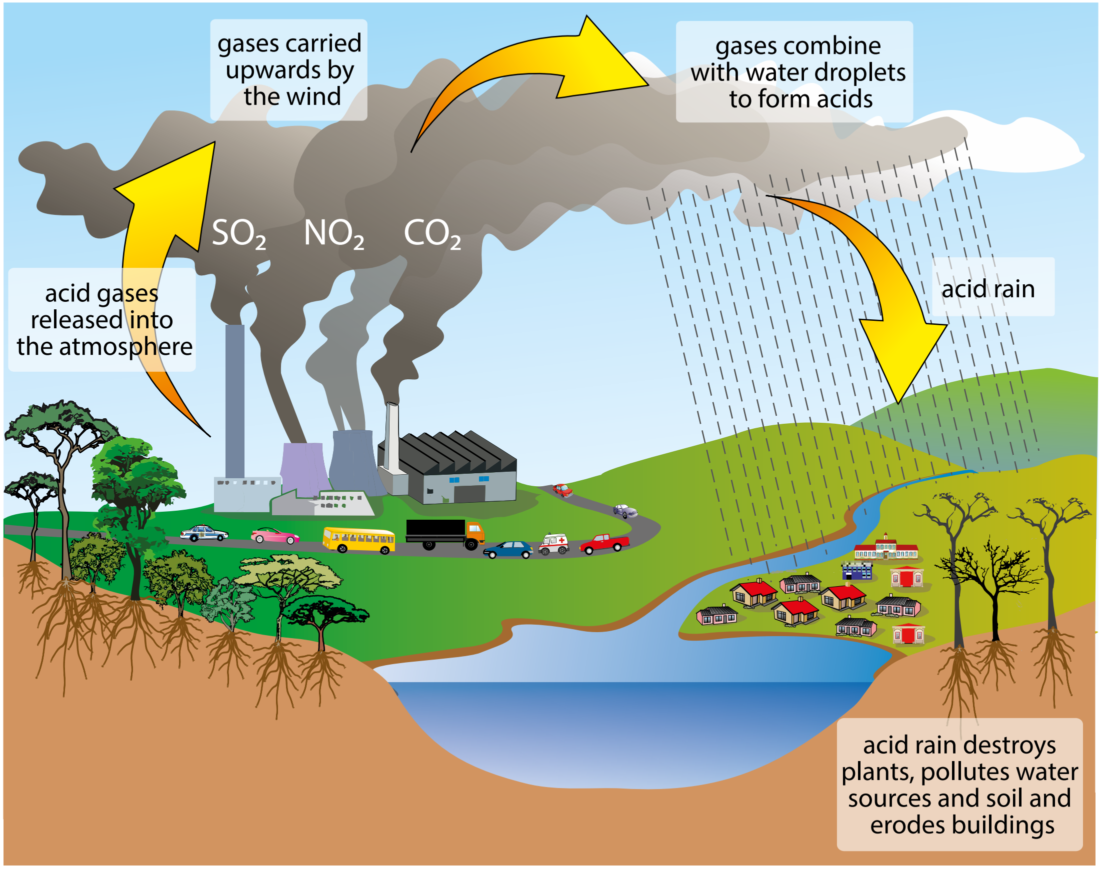


Figure 11.8: Formation of acid rain

Questions

1. Which three gases are shown in the diagram that contribute to the formation of acid rain? Write their names and formulae.
2. What are some of the sources of these gases? You can do some extra reading about this to help you answer this question.
3. Write the equations for how two of these gases which you have learnt about react with the water in the atmosphere to form acids.
4. What are the names of these two acids?
5. What are some of the environmental impacts of acid rain? Study the diagram for some clues and do some extra reading.
6. Acid rain can also damage buildings as it 'eats away' the stone. What property of acids allows it to do this?
7. Factories used to have quite short funnels to let out the smoke, but it was found that this caused many problems in the local towns and cities near the factory as the gases would combine with water in the immediate environment to cause acid rain. Factories then started to build much

higher smoke funnels so that the smoke was let out high enough to be blown further away.
Do you think this is an efficient way to help reduce acid rain?
Explain your answer.

8. Do some research to find out about the possible ways to prevent or minimise the formation of acid rain. Write a paragraph to summarise these methods in your exercise books.

We have now learnt about non-metal oxides, but what about metal oxides?

What kind of solutions do they form in water? We will find out more about them and other metal compounds in the next section.

11.2 The general reaction of an acid with a metal oxide

New word

- metal oxide

In the previous section we learnt about two classes of oxides, namely metal oxides and non-metal oxides. Here is what we know about them so far:

- Metal oxides are formed from the reaction between a metal and oxygen.
- Metal oxides are basic. When we dissolve them in water, they form solutions with pH values above 7.
- Non-metal oxides are formed from the reaction between a non-metal and oxygen. Non-metal oxides are acidic. When they dissolve in water, they form solutions with pH values below 7.
- Here is the same summary, in table form, with some examples added:

Metal oxides	Non-metal oxides
metal + oxygen → metal oxide	non-metal + oxygen → non-metal oxide
basic	acidic
pH > 7	pH < 7
Examples: Li_2O, Na_2O, MgO, CaO	Examples: CO_2, SO_2, NO_2, P_2O_5

In this section, we are going to learn about the reactions between metal oxides and acids.

Investigation: The reaction between magnesium oxide and hydrochloric acid

Aim

- To test whether a solution of magnesium oxide in water is acidic, basic or neutral; and
- to determine whether the reaction between an aqueous solution of magnesium oxide and hydrochloric acid is a neutralisation reaction.

Investigative question(s)

What are the questions you hope to answer with this investigation? Write them in your exercise books. There are a few words to start you off.

1. When magnesium oxide is dissolved in water, will the resulting solution …?
2. When a solution of magnesium oxide is treated with hydrochloric acid, will the pH of the mixture …?

Overview of the investigation

1. We will measure the pH of a solution of magnesium oxide (MgO) with universal indicator paper to confirm whether it is acidic or basic. Within what range do you expect the pH of the magnesium oxide solution to fall?
2. We will add hydrochloric acid (HCl) to the magnesium oxide solution in small portions and measure the pH after each portion has been added. What changes do you expect to observe – will the pH increase, decrease, or stay the same?

Hypothesis

What are your predictions? Your hypothesis should be a prediction of the finding(s) of the investigation. You should write it in the form of a possible answer to the investigative question(s). Here are a few words to start you off:

1. When magnesium oxide is dissolved in water, the resulting solution will…
2. When a solution of magnesium oxide is treated with hydrochloric acid, the pH of the mixture will…

Materials

- magnesium oxide powder
- water
- universal indicator paper (cut into 1 cm strips)
- white tile or sheet of white printer paper
- glass rod (or plastic straw)
- test tube
- dropper
- diuted hydrochloric acid (HCl) solution

Method

1. Prepare the universal indicator paper by neatly placing five 1 cm pieces in a column on the white tile or sheet of printer paper.
2. Place a small quantity (the size of a match head) of the magnesium oxide in a test tube.
3. Add approximately 2 ml of tap water to dissolve most of the magnesium oxide.
4. Use the glass rod (or plastic straw) to stir the solution until most the magnesium oxide has dissolved. We will be calling this the test solution from now on.

5. Transfer one drop of the test solution to the first piece of universal indicator paper.
6. Compare the colour of the paper with the colour guide to find the pH of the solution.
7. Record this pH in the table you prepared beforehand.
8. Add 10 drops of the hydrochloric acid solution to the test solution. Stir it gently and transfer another drop of the solution to a fresh strip of the universal indicator.
9. Read the pH of the solution off the colour guide and record it in your table.
10. Repeat steps 3 and 4 until the pH of the test solution drops below 7. You may need more than 5 pieces of universal indicator paper.

Results

1. Present your results in a table. You should prepare this beforehand. Use appropriate headings for your table. 'Number of drops of HCl added' and, 'Colour of the universal indicator paper', and 'pH of the test solution' are suggested headings for your columns.
2. Draw a graph of your results. Here are some hints to help you decide which variable to put on which axis:
   a) What is your independent variable? (Which variable did you change?)
   This goes on the x-axis.
   b) What is your dependent variable? (Which variable did you measure?)
   This goes on the y-axis.

Conclusions

What conclusions can be made from the results of your investigation? Reflect on your hypothesis, but change it to reflect your findings if they are different from what you predicted earlier.

Were you able to confirm or reject your hypotheses?

Now that we have investigated a reaction between a metal oxide (MgO) and an acid (HCl), we can write an equation for the reaction. We will begin by writing a general equation and end with one that matches the reaction that we have just investigated.

General equation for the reaction of an acid with a metal oxide

Can you remember learning that an acid-base reaction is an exchange reaction?

We learnt that:

- The acid contributes H towards making a water molecule;
- The base contributes O or OH towards making a water molecule; and
- Whatever is left of the acid and the base after making an H_2O molecule combines to form a salt.

The general word equation for the reaction between an acid and a base is:

acid + base → salt + water

Since the base in our reaction is a metal oxide we can write:

acid + metal oxide → salt + water

This is the general word equation for the reaction between an acid and a metal oxide. The type of salt that forms will depend on the specific acid and metal oxide which were used in the reaction.

Equations for the reaction between magnesium oxide and hydrochloric acid

Now we are going to learn how to write equations for our actual reaction.

ACTIVITY Writing the chemical equation

The following steps will guide you:

1. The acid of our reaction was hydrochloric acid. Write its chemical formula.
2. What is the name and formula of the metal oxide we used?
3. Now, let's try to predict the products of the reaction. We know that water will be one of the products.
4. Write what remains of the base (MgO) after we have taken away the O (to make water).
5. Write what remains of the acid (HCl) after we have taken away the H (to make water). (Remember we need two H to make one H_2O).
6. Now put the two remaining fragments together. Place the metal first and remember that 2 HCl will leave 2 Cl after the 2 H has been given to O to make water. One Mg and 2 Cl makes …
   Now, let's put it all together, in the following order.
   $2\ HCl + MgO \rightarrow MgCl_2 + H_2O$
7. Let's check quickly if the reaction is balanced.
   a) How many H atoms on the left-hand side and on the right-hand side? Are they balanced?
   b) How many Cl atoms on the left-hand side and on the right-hand side? Are they balanced?
   c) How many O atoms on the left-hand side and on the right-hand side? Are they balanced?

   Since the numbers of each type of atom is the same on either side of the equation, we can confirm that it is balanced.

Finally, let's use the chemical equation to write a word equation for the reaction:

hydrochloric acid + magnesium oxide → magnesium chloride + water

In the next section we are going to look at the reactions between acids and metal hydroxides.

11.3 The general reaction of an acid with a metal hydroxide

Keywords

- metal hydroxide

We will start this section with an investigation to illustrate the reaction between an acid and a **metal hydroxide**.

Investigation: The reaction between sodium hydroxide and hydrochloric acid

Aim

- Test whether a solution of sodium hydroxide in water is acidic, basic or neutral; and
- determine whether the reaction between an aqueous solution of sodium hydroxide and hydrochloric acid is a **neutralisation reaction**.

Investigative question(s)

What are the questions you hope to answer with this investigation? Write them in your exercise books. You may use the previous investigation (of the reaction between magnesium oxide and hydrochloric acid) as a guideline.

Overview of the investigation

1. We will measure the pH of a solution of sodium hydroxide ($NaOH$) with universal indicator paper to confirm whether it is acidic or basic. Within what range do you expect the pH of the magnesium oxide solution to fall?
2. We will add hydrochloric acid (HCl) to the sodium hydroxide solution in small portions and measure the pH after each portion has been added.

What changes do you expect to observe – will the pH increase, decrease, or stay the same?

Hypothesis

What are your predictions? Your hypothesis should be a prediction of the finding(s) of the investigation. You should write it in the form of a possible answer to the investigative question(s). If you are unsure, check the previous investigation.

Materials

- sodium hydroxide solution (0,1 M)
- universal indicator paper (cut into 1 cm strips)
- white tile or sheet of white printer paper
- glass rod (or plastic straw)
- test tube
- plastic syringe
- hydrochloric acid (HCl) solution (0,1 M)

Method

1. Prepare the universal indicator paper by neatly placing five 1 cm pieces in a column on the white tile or sheet of printer paper.
2. Use the syringe to transfer 2 ml of the sodium hydroxide solution into the test tube or small glass beaker. We will be calling this the test solution from now on.
3. Rinse the syringe very thoroughly with water and dry it out with a clean tissue. Now fill it with hydrochloric acid solution and set it aside.
4. Transfer one drop of the sodium hydroxide (test solution) to the first piece of universal indicator paper.
5. Compare the colour of the paper with the colour guide to find the pH of the sodium hydroxide solution. Record this pH in your results table.
6. Add 0,5 ml of the hydrochloric acid solution from the syringe to the test solution. Stir it gently with the glass rod or straw and transfer another drop of the test solution to a fresh strip of the universal indicator paper.
7. Read the pH of the solution off the colour guide and record it in your results table.
8. Repeat steps 6 and 7 until the pH of the test solution reaches approximately 7.
9. How much of the hydrochloric acid solution have you used? Write the volume in your exercise books.
10. If you are quite sure that all the base has been neutralised by the acid (the pH should be 7 and the universal indicator paper should have turned green), pour the test solution into a small glass beaker and leave it in the window sill for a few days. Remember to come back to it later to see what has happened to it.

Results

1. Present your results in a neat table. Use appropriate headings for your table.
   The following are suggested headings for your columns:
   - Volume of HCl added
   - Colour of the universal indicator paper
   - pH of the test solution.
2. Draw a graph of your results.
   a) What is your independent variable? (Which variable did you change?)
   b) What is your dependent variable? (Which variable did you measure?)

Conclusion

What conclusion can be drawn from the results of your investigation? Here you can rewrite your hypothesis, but change it to reflect your findings if they are different from what you predicted earlier.

Were you able to confirm or reject your hypotheses?

Now that we have investigated a reaction between a metal oxide (NaOH) and an acid (HCl), we can write an equation for the reaction. We will begin by writing a general equation and end with one that matches the reaction that we have just investigated.

General equation for the reaction of an acid with a metal hydroxide

You learnt that an acid–base reaction can be represented by the following general word equation.

acid + base → salt + water

The base in our reaction was a metal hydroxide, so the general word equation becomes:

acid + <u>metal hydroxide</u> → salt + water

This is the general word equation for the reaction between an acid and a metal hydroxide. The type of salt that forms will depend on the specific acid and metal hydroxide which were used in the reaction.

Equations for the reaction between sodium hydroxide and hydrochloric acid

Now we are going to learn how to write equations for our actual reaction.

ACTIVITY Writing the chemical equation

The following steps will guide you

1. The acid of our reaction was hydrochloric acid. Write its chemical formula.
2. What is the name and formula of the metal hydroxide we used?
3. Now, let's try to predict the products of the reaction. We know that water will be one of the products.
4. Write what remains of the base after we have taken away the OH to make water.
5. Write what remains of the acid after we have taken away the H to make water. Remember we need two H to make one H_2O, but NaOH has already contributed one O and one H. Now put the two fragments together. Place the metal from the base first and the non-metal from the acid. One Na and one Cl makes...
6. Now, let's put it all together, in the following order:
   Acid + metal hydroxide → salt + water
7. Let's check quickly if the reaction is balanced.
   a) How many H atoms on the left and on the right? Are they balanced?
   b) How many Cl atoms on the left and on the right? Are they balanced?
   c) How many O atoms on the left and on the right? Are they balanced?
8. Once you have performed this reaction and you are left with a neutral solution, you decide you want to recover the sodium chloride (table salt). How will you do this?

Finally, let's use the chemical equation to write a word equation for the reaction:

hydrochloric acid + sodium hydroxide → sodium chloride + water

In the next section we are going to look at the reactions between acids and metal carbonates.

Keywords

- metal carbonate

11.4 The general reaction of an acid with a metal carbonate

In this section we will investigate the reaction between an acid and a **metal carbonate**.

Figure 11.9: Blackboard chalk is calcium carbonate, a metal carbonate.

Investigation: The reaction between calcium carbonate (chalk) and hydrochloric acid

Aim

- test whether calcium carbonate is acidic or basic;
- determine whether the reaction between calcium carbonate and hydrochloric acid is a neutralisation reaction; and
- determine the products of the reaction between calcium carbonate and hydrochloric acid.

Investigative questions

What are the questions you hope to answer with this investigation? Write them in your exercise books. You may use your previous investigations as a guideline.

Overview of the investigation

1. We will measure the pH of a suspension of calcium carbonate ($CaCO_3$) with universal indicator paper to confirm whether it is acidic or basic.
   Within what range do you expect the pH of the calcium carbonate to fall?
2. We will add hydrochloric acid (HCl) to the calcium carbonate in small portions and measure the pH after each portion has been added. What changes do you expect to observe? Will the pH increase, decrease, or stay the same?

Hypothesis

What are your predictions? Your hypothesis should be a prediction of the finding(s) of the investigation. You should write it in the form of a possible answer to the investigative question(s). If you are unsure, check the previous investigation.

Materials

- chalk dust (calcium carbonate) suspended in a small quantity of water.
- universal indicator paper (cut into 1 cm strips)

- white tile or sheet of white printer paper
- glass rod or plastic straw
- test tube or small glass beaker
- plastic syringe (2,5 cm capacity) or dropper
- diluted hydrochloric acid (HCl) solution

Method

1. Prepare the universal indicator paper by neatly placing five 1 cm pieces in a column on the white tile or sheet of printer paper.
2. Place approximately 2 ml of the calcium carbonate suspension into the test tube or small glass beaker. We will be calling this the test solution from now on.
3. Rinse the syringe very thoroughly with water and dry it out with a clean tissue. Now fill it with hydrochloric acid solution and set it aside.
4. Transfer one drop of the calcium carbonate (test solution) to the first piece of universal indicator paper.
5. Compare the colour of the paper with the colour guide below, to find the pH of the calcium carbonate solution. Record this pH in your results table.
6. Add 0,5 cm of the hydrochloric acid solution from the syringe to the test solution. Watch very carefully what happens. Do you see anything interesting? (Hint: Look for bubbles!) Stir the test solution gently with the glass rod and transfer another drop of it to a fresh strip of the universal indicator.
7. Read the pH of the solution off the colour guide and record it in your table.
8. Repeat steps 6 and 7 until the pH of the test solution reaches approximately 7. How much of the hydrochloric acid solution have you used? Write the volume in your exercise books.
9. Your teacher will repeat the experiment as a demonstration and will collect the gas that forms during the reaction, for testing with clear lime water.
10. Can you remember which gas we are testing for with clear limewater? Write its name and formula in your exercise books.

Results

1. Present your results in a neat table. Use appropriate headings for your table. Suggested headings for your columns are as follows:
   - Volume of HCl added
   - Colour of the universal indicator paper
   - pH of the test solution
2. Draw a graph of your results.
   a) What is your independent variable? (Which variable did you change?)
   b) What is your dependent variable? (Which variable did you measure?)

Conclusions

What conclusions can be made from the results of your investigation? Reflect your findings if they are different from what you predicted earlier.

Were you able to confirm or reject your hypothesis?

Now that we have investigated a reaction between a metal carbonate ($CaCO_3$) and an acid (HCl), we can write an equation for the reaction. We will begin by writing a general equation and end with one that matches the reaction that we have just investigated.

General equation for the reaction of an acid with a metal carbonate

The general equation for the reaction between an acid and a base is as follows:

acid + base → salt + water

If we replace 'base' with 'metal carbonate', we get:

acid + metal carbonate → salt + water

But wait, there was a third product in our reaction! Can you remember what it was? (Hint: Bubbles formed, so it was a gas.)

We need to make it clear that CO_2 was a product of the reaction, so the correct general word equation would be:

acid + metal carbonate → salt + water + carbon dioxide

The type of salt that forms will depend on the specific acid and metal carbonate which were used in the reaction.

Equations for the reaction between calcium carbonate and hydrochloric acid

Now we are going to learn how to write equations for our actual reaction.

ACTIVITY Writing the chemical equation

The following steps will guide you

1. The acid of our reaction was hydrochloric acid. Can you write its chemical formula?
2. What is the name and formula of the metal carbonate we used?
3. Now, let's try to predict the products of the reaction. We know that water and carbon dioxide will be two of the products.
4. Write what remains of the base after we have taken away the CO to make CO_2 and leave one O_2 to make water.

5. Write what remains of the acid after we have taken away the H to make water. Remember we need two H to make one H_2O and $CaCO_3$ has contributed only one O_2.
6. Now put the two fragments together. Place the metal from the base first and the non-metal from the acid. Ca and 2 Cl makes ...
7. Now, let's put it all together, first the reactants, then the products:
   $2\ HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$
8. Let's check quickly if the reaction is balanced.
   a) How many H on the left and right? Are they balanced?
   b) How many Cl on the left and right? Are they balanced?
   c) How many O on the left and right? Are they balanced?
   d) How many C on the left and right? Are they balanced?

Finally, let's use the chemical equation to write a word equation for the reaction:

hydrochloric acid + calcium carbonate → calcium chloride + water + carbon dioxide

Applications for calcium carbonate

Calcium carbonate is found in many places outside of the laboratory. It is found in different types of rocks around the world, for example limestone, chalk and marble.

Figure 11.10: The Cango Caves near Oudtshoorn, South Africa, are situated in a limestone ridge and contain spectacular limestone formations. Such caves are the result of water high in carbonic acid acting upon limestone deposits in ancient rock layers.

Did you know?

Dark green leafy vegetables such as broccoli, kale and cabbage are a dietary source of calcium carbonate, providing the body with calcium. You can also take calcium carbonate in the form of tablet supplements.

Calcium carbonate is also the main part of shells of various marine organisms, snails, pearls, oysters and bird eggshells. It is also found in the exoskeletons of crustaceans (such as crabs, prawns and lobsters).

Calcium carbonate also has many applications. In industry, the main application is in construction as it is used in various building materials and in cement.

Figure 11.11: Chunks of calcium carbonate from various shells.

Calcium carbonate is used in many adhesives, paints and in ceramics. It is also used in swimming pools to adjust the pH. When do you think it would be added? If the pool was too acidic and you wanted to make it more basic, or if the pool was too basic and you wanted to make it more acidic?

Calcium carbonate is also used in agriculture in the form of lime powder. Agricultural lime is made by grinding up limestone or chalk. It is added to the soil if the soil is too acidic to increase the pH. It also provides plants with a source of calcium.

Figure 11.12: This tractor is busy depositing agricultural lime onto a field. This is called liming.

In this unit we have investigated a number of reactions of acids with bases. We have learnt to write word equations for these reactions and practised converting between word and balanced chemical equations.

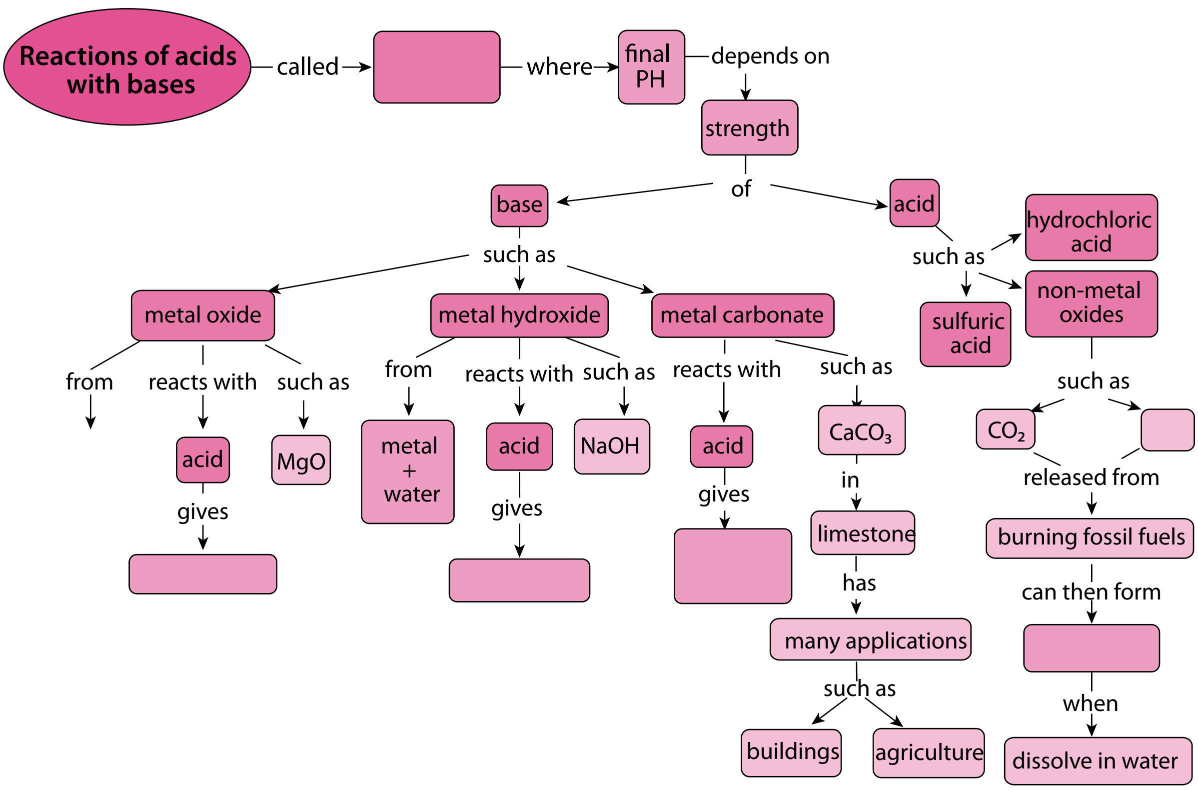
Summary

Key concepts

- The reaction of an acid with a base is called a neutralisation reaction.
- When an acid (pH < 7) is added to a base (pH > 7), the pH of the resulting mixture will lie somewhere between that of the acid and the base. Even though the acid and base will be neutralised, the resulting solution will not necessarily be neutral.
- Some common laboratory acids are sulfuric acid (H_2SO_4), nitric acid (HNO_3) and hydrochloric acid (HCl).
- Non-metal oxides tend to form acidic solutions when they dissolve in water. These solutions will have pH values below 7.
- Metal oxides, metal hydroxides and metal carbonates form basic solutions in water; these will have pH values above 7.
- When a metal oxide or a metal hydroxide reacts with an acid, a salt and water form as products.
- When a metal carbonate reacts with an acid, a salt, water and carbon dioxide form as products.
- The general word equations for the reactions of this unit are the following:
  - acid + metal oxide → salt + water
  - acid + metal hydroxide → salt + water
  - acid + metal carbonate → salt + water + carbon dioxide

Concept map

Complete the concept map provided by your teacher by filling in the blank spaces.



Revision

1. Fill in the missing words in these sentences. Write the complete sentences in your exercise books. [10]
   a) To know if something is an acid or a base, we measure its ___________________
   b) The name of the laboratory acid with the formula H_2SO_4 is ___________________
   c) The formula of the laboratory acid named hydrochloric acid is ___________________
   d) When a metal oxide reacts with an ___________________, a salt and water will be formed.
   e) When a metal hydroxide reacts with an acid, a salt and ___________________ will be formed.
   f) When a metal carbonate reacts with an acid, a salt, water and ___________________will be formed.
   g) Metal oxides, metal hydroxides and metal carbonates all dissolve in water, forming ___________________ solutions. This means the solutions will have pH values ___________________than 7.
   h) The reaction of an acid with a base is called a ___________________reaction.
   i) Non-metal oxides tend to form ___________________ solutions when they dissolve in water.

2. Write a short paragraph (3 or more sentences) to explain what you understand each of the following terms to mean, in your own words. [2 × 3 = 6]
   a) neutralisation
   b) acid rain

3. Copy the following tables in your exercise books. Complete the tables for each of the reactions by providing the missing equations.
   a) The reaction between hydrochloric acid and magnesium oxide. [4]

Word equation	
Chemical equation	
General equation	acid + metal oxide → salt + water

   b) The reaction between hydrochloric acid and sodium hydroxide. [6]

Word equation	
Chemical equation	
General equation	

   c) The reaction between hydrochloric acid and calcium carbonate. [4]

Word equation	
Chemical equation	$2\,HCl + CaCO_3 \rightarrow CaCl_2 + H_2O + CO_2$
General equation	

d) The reaction between hydrochloric acid and magnesium hydroxide. [4]

Word equation	
Chemical equation	$2\ HCl + Mg(OH)_2 \rightarrow MgCl_2 + 2\ H_2O$
General equation	

e) The reaction between hydrochloric acid and calcium oxide. [4]

Word equation	
Chemical equation	$2\ HCl + CaO \rightarrow CaCl_2 + H_2O$
General equation	

f) The reaction between hydrochloric acid and potassium hydroxide. [6]

Word equation	
Chemical equation	
General equation	

g) The reaction between hydrochloric acid and sodium carbonate. [4]

Word equation	hydrochloric acid + sodium carbonate → sodium chloride + water + carbon dioxide
Chemical equation	
General equation	

Total [48 marks]

12 Reactions of acids with metals

Key questions

- What do we get when a metal reacts with an acid?
- What is the general equation for the reaction between a metal and an acid?
- How do we write the word equation and the balanced chemical equation?
- How can we test for the presence of hydrogen gas?

Keywords

- diatomic
- density
- characteristic
- presence
- chemist
- pharmacist

12.1 The reaction of an acid with a metal

In the previous unit we learnt about the reactions of acids with a variety of bases: metal oxides, metal hydroxides and metal carbonates. We learnt how to write general equations, word equations and chemical equations for those reactions.

In this unit we will investigate one final type of reaction, namely the reaction between an acid and a metal.

First, we will do an investigation to observe the reaction and then we will write equations to represent it. Before we do this, however, we have to take a quick detour to learn something interesting about hydrogen gas.

ACTIVITY Testing for hydrogen gas

1. What do you know about hydrogen gas? Perhaps you know its formula? Write it in your exercise books.
2. Hydrogen gas is a diatomic gas. What does this mean?
3. What do you know about the position of hydrogen in the periodic table? Write what you know in your exercise books.
4. The position of hydrogen in the periodic table tells us that it is the lightest of all the elements. It has the smallest atomic mass. Because the element hydrogen is a gas (even though it is a diatomic one), it has one of the lowest densities of any substance.

Figure 12.1: This man is about to launch a weather balloon filled with hydrogen gas. It will float upwards to collect information about the weather in Antarctica.

Can you remember what density means? Write your own definition in your exercise books.

When hydrogen gas is released in a reaction it will immediately rise up, because hydrogen is less dense than air. If you filled a balloon with hydrogen, it would float up and you would need to tie a string to it to prevent it from floating away!

Another interesting thing about hydrogen is that it reacts explosively with oxygen if you bring a flame near it.

You may remember learning about this in Unit 9 about the reactions of non-metals with oxygen. The reaction between a large quantity of hydrogen and oxygen in the air produces a beautiful orange fireball and a very loud boom! Do you remember seeing the following diagram?

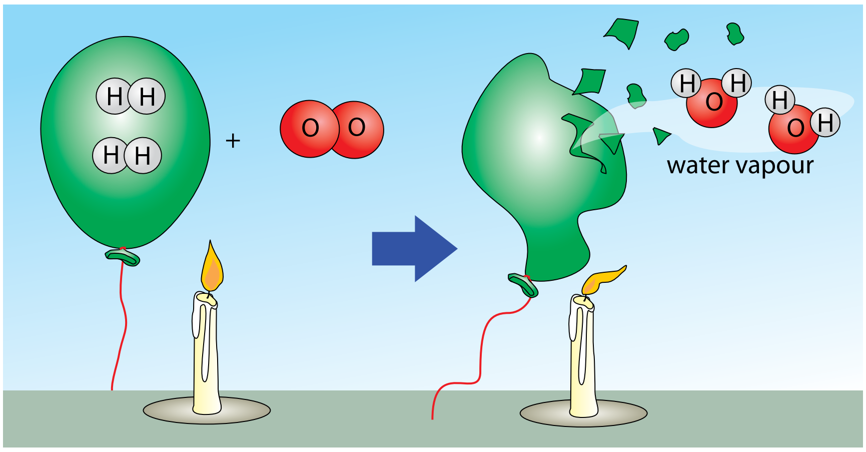


Figure 12.2

5. Write the balanced equation for the reaction between hydrogen gas and oxygen in your exercise books.
The reaction between a tiny amount of hydrogen and oxygen in the air produces a **characteristic** 'pop' sound and this serves as a test for the **presence** of hydrogen. You can watch the short video clip in the visit box in the margin to see this 'pop'.

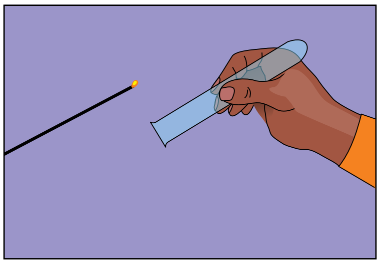


Figure 12.3: When a glowing splint is put into a test tube containing hydrogen gas …

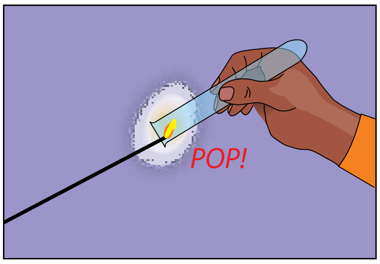


Figure 12.4: it makes a 'pop' sound.

Did you know?

The name 'Hydrogen' comes from the Greek words 'hydro' and 'gen', which means 'water generator'.

Let's now investigate the reaction between an acid and a metal. You should listen carefully for this 'pop' sound during the investigation. If you hear it, it will signal the presence of hydrogen gas!

Investigation — The reaction between magnesium and hydrochloric acid

Aim

- Observe the reaction between hydrochloric acid and magnesium; and
- identify the gaseous product of the reaction between hydrochloric acid and magnesium.

Your teacher will demonstrate the reaction between magnesium and hydrochloric acid, while you make observations. Remember to watch carefully and take detailed notes.

Investigative question

What question(s) do you hope to answer with this investigation?

Hypothesis

What do you predict will happen? Your hypothesis should be a prediction of the finding(s) of the investigation. You should write it in the form of a possible answer to the investigative question(s).

Materials

- magnesium ribbon (cut into smallish pieces)
- concentrated hydrochloric acid (HCl) solution
- large test tube
- retort stand with clamp
- rubber stopper with short length of glass tubing pushed through it
- silicone or rubber tubing (or a glass delivery tube as shown in the set-up below)
- shallow dish filled with soapy water (made by mixing a few teaspoons of dishwashing liquid with water)

Method

1. Use a piece of universal indicator paper to test the pH of the hydrochloric acid solution. Record its pH.
2. Set up the experiment as shown in the following diagram. Ensure that the end of the delivery tube is below the surface of the soap solution in the dish.

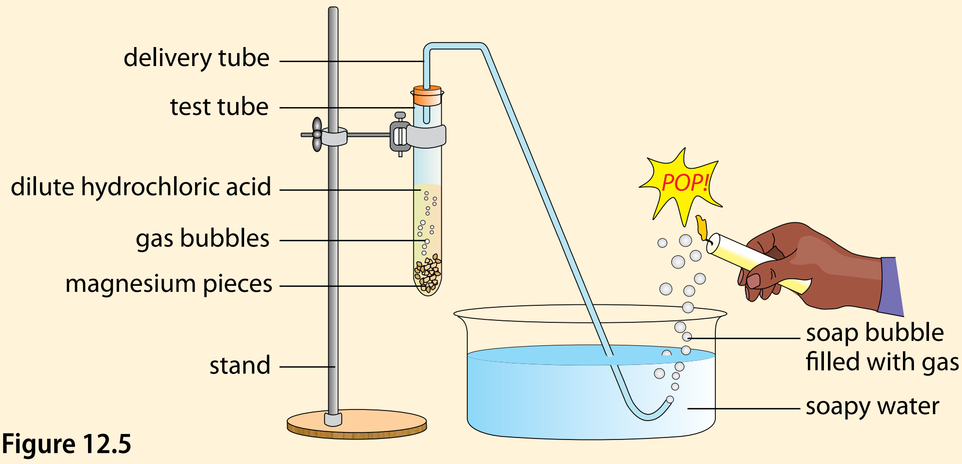


Figure 12.5

3. Place approximately 1 g of the magnesium pieces in the test tube, but do not add the hydrochloric acid until everything else is ready to be assembled.
4. Add approximately 40 ml of hydrochloric acid and immediately place the stopper on the test tube. The first few bubbles of gas that are released from the end of the delivery tube will be air.
5. When the soap bubbles start to float up, hold a burning candle to them and listen carefully for the sound they make when they pop. Do not hold the candle to the end of the delivery tube.
6. When the magnesium stops reacting and no further hydrogen bubbles are released, extinguish the candle and set it aside.
7. Disassemble the experiment and test the pH of the reaction mixture. Record the pH value.

Results and observations

Record your results in the following table:

pH of the hydrochloric acid before the reaction	
pH of the mixture after the reaction	

Write down the observations you make during the investigation in your exercise books.

Conclusions

What conclusions can be made from the results of your investigation? Reflect on our hypothesis, but change it to reflect your findings if they are different from what you predicted earlier.

Questions

1. What did you observe in the test tube when the magnesium and hydrochloric acid were mixed?
2. What did you observe at the end of the gas delivery tube after the magnesium and hydrochloric acid were mixed?
3. Why do you think the soap bubbles floated upwards?
4. Which gas do you think was produced by the reaction? Write its name and formula in your exercise books. What makes you think it was this gas?
5. What happened to the pH of the hydrochloric acid solution during the reaction?
6. What does the increase in pH mean?
7. Were you able to confirm or reject your hypothesis?

In our investigation hydrochloric acid reacted with magnesium (a metal). Our next task is to write an equation for the reaction. We will begin by writing a general equation and end with one that matches the reaction that we have just investigated.

General equation for the reaction of an acid with a metal

The general word equation for the reaction between an acid and a metal is:

acid + metal → salt + hydrogen gas

The type of salt that forms will depend on the specific metal and acid which are used in the reaction.

Equations for the reaction between magnesium and hydrochloric acid

Now let's write equations for our actual reaction from the last investigation.

ACTIVITY Writing the chemical equation

The following steps will guide you

1. The acid of our reaction was hydrochloric acid. Can you write its chemical formula?
2. What is the name and formula of the metal we used?
3. Now, let's try to predict the products of the reaction. We know that hydrogen gas will be one of the products. Write the chemical formula for hydrogen gas.
4. Write what remains of the acid (HCl) after we have taken away the H to make H_2. (Remember we need two H to make one H_2).
5. The two Cl and the Mg are exactly what are needed to make magnesium chloride. Write the formula in your exercise books.
6. Now, let's write the reaction; first the reactants, then the products:

   $$\mathbf{2\,HCl + Mg \rightarrow MgCl_2 + H_2}$$

   Let's check quickly if the reaction is balanced.
   a) How many H on the left and right? Are they balanced?
   b) How many Cl on the left and right? Are they balanced?
   c) How many Mg on the left and right? Are they balanced?

   Since the numbers of each type of atom is the same on either side of the equation, we can confirm that it is balanced.
7. Finally, let's use the chemical equation to write a word equation for the reaction.

Chemist or pharmacist?

When people hear that someone is a 'chemist', they often confuse this with being a 'pharmacist'. In some countries the terms 'chemist' and 'pharmacist' are even used to describe the same kind of person. In South Africa the two words have different meanings. But what is the difference between being a chemist and being a pharmacist?

Look up these careers to identify the main difference between a chemist and a pharmacist and summarise them in your books.

Figure 12.6: Two chemists working in a laboratory.

Figure 12.7: A pharmacist in his dispensary.

ACTIVITY Other careers in chemistry

Instructions

1. Below is a list of different careers that all use chemistry in some way. Have a look through the list and then select the five careers you find most interesting.
2. Do an internet search to find out what each career is.
3. Write a one-line description of this career in your exercise books.
4. If there is a career that really interests you, draw a smiley face next to it and be sure to do some extra reading around the topic and where chemistry may take you! Find out what level of chemistry you will need for this particular career.
5. There are many other careers besides the ones listed here which use chemistry in some way, so if you know of something else which is not listed here and it interests you, follow your curiosity and discover the possibilities!

Some careers involving chemistry

- Agricultural chemistry
- Biochemistry
- Biotechnology
- Environmental chemistry
- Chemical education/teaching
- Chemistry researcher
- Forensic science
- Food science/technology
- Geneticist
- Materials science
- Geochemistry
- Medicine and medicinal chemistry
- Oil and petroleum industry
- Organic chemistry
- Oceanography
- Patent law
- Pharmaceuticals
- Space exploration
- Zoology

In this unit we have studied the reaction of hydrochloric acid with magnesium, as an example of a reaction between an acid and a metal.

Did you know?

Great scientists (including chemists) are observant, curious, patient and eager to learn more about their field every day.

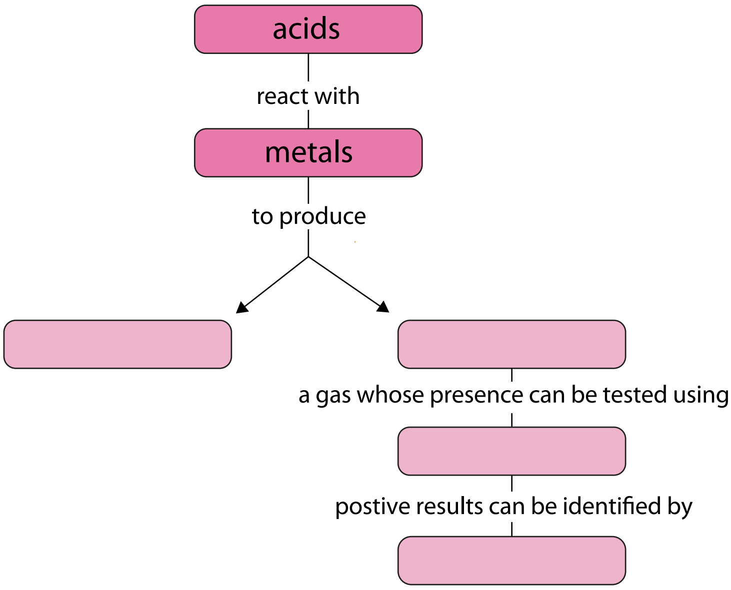
Summary

Key concepts

- An acid will react with a metal to form a salt and hydrogen gas.
- The general word equation for the reaction between an acid and a metal is as follows:
  - acid + metal → salt + hydrogen

Concept map

This was quite a short unit, so the concept map has been left blank for you to do your own. Be sure to include something about the test for hydrogen.



Revision

1. Fill in the missing words in these sentences. Write the word in your exercise books. [3]
   a) When an acid reacts with a metal, a salt and ___________ gas forms.
   b) A molecule that consists of two atoms bonded together is called a ___________ molecule.
   c) The scientific quantity represented by the mass of a substance in a given volume is called the ___________ of that substance.

2. Write a short paragraph (2 sentences or more) to explain why a balloon filled with hydrogen will float upwards. [2]

3. Imagine you are carrying out a reaction and you expect one of the products that will form to be hydrogen. Write a short paragraph (2 sentences or more) to describe how you would confirm the presence of hydrogen gas. [2]

4. When an acid reacts with a metal, do you think the pH of the solution will increase, decrease, or stay the same? Motivate your answer briefly. [3]

5. Copy and complete the following table in your exercise books by providing the missing equations for the reaction between hydrochloric acid and magnesium. [6]

Word equation	
Balanced chemical equation	
General equation	

6. Copy and complete the following table in your exercise books by providing the missing equations for the reaction between hydrochloric acid and zinc. [4]

Word equation	
Balanced chemical equation	$2\ HCl + Zn \rightarrow ZnCl_2 + H_2$
General equation	

7. We have looked at many different chemical reactions this term. As a summary, copy and complete the following table in your exercise books by giving the general equations in words for each of the chemical reactions in the second column, and provide an example for each type as a balanced chemical equation in the third column. [18]

Type of chemical reaction	General word equation	Example (balanced equation)
metals with oxygen		
non-metals with oxygen		
acids with metal oxides		
acids with metal hydroxides		
acids with metal carbonates		
acids with metals		

Total [38 marks]

Glossary

acid rain rainwater that is unusually acidic as a result of dissolved non-metal oxides that have entered the atmosphere

acidity this word is related to the word 'acid', and so is the word 'acidic'; a substance is strongly acidic when it has a high (degree of) acidity

alternative different

atomic number a unique number that represents a given element, and shows its position on the periodic table; the number of protons found in the nucleus

balanced a balanced equation reaction has the same numbers of atoms of a particular type on opposite sides of the reaction equation

barrier a fence or other obstacle that keeps things apart

bond a force between atoms in a compound, holding them together

camera flash a device (usually attached to the camera) that provides a quick burst of light at the instant that the photo is taken

characteristic a quality or feature of an object or item; for example, one of the characteristics of an acid is that it is corrosive

chemical bond a special attractive force that holds the atoms in a molecule together

chemical equation an equation that describes a chemical reaction using the chemical formulae of the compounds involved in the reaction

chemical formula a combination of element symbols that shows the types and number of atoms in one molecule of a given compound; a unique string of symbols (letters and numbers) that represents a chemical compound

chemical reaction a process in which atoms in substances, called reactants, are rearranged to form new substances, called products

chemist a person who has studied chemistry and uses this knowledge to do his/her job

chromed metal metal that is covered by a thin layer of chromium

coefficient a number that is placed in front of a chemical formula in a reaction equation; it shows the number of molecules of that type taking part in the reaction, for example 2 Mg

collide to bump into something

combustion a type of chemical reaction where a substance and oxygen react during burning to form a new product

compound a pure substance in which atoms of two or more different chemical elements are bonded in a fixed ratio

corrosion the gradual destruction of materials (usually metals) by chemical reaction with substances in the environment

corrosive a corrosive substance is something that causes corrosion; substances that are corrosive can cause burns on the skin and damage to certain surfaces

crystal lattice in some compounds, the atoms are arranged in a regular pattern in a fixed ratio to form a lattice structure; a lattice looks like a mesh or trellis

density the mass of a substance in a given space (volume)

detour to take a roundabout route, either to make a visit along the way, or to avoid something

diatomic molecule consists of two atoms; H_2, O_2, F_2, Cl_2, Br_2, and I_2 are all examples of elements that consist of diatomic molecules

dioxide a compound that contains two oxygen atoms in its chemical formula; examples are carbon dioxide (CO_2) and sulfur dioxide (SO_2)

electrons the smallest of the three types of sub-atomic particles; they are negatively charged and are located outside the atomic nucleus

element a pure substance that consists of only one type of atom throughout

exchange reaction a reaction in which the reactants break up into fragments that are then exchanged, or swapped around

exposed when a material is exposed, it is uncovered or unprotected (in this case from oxygen that will react with it)

fossil fuel a fuel that was formed from the prehistoric remains of plant and animal life (fossils); it usually has to be extracted from the earth; examples include oil, coal and natural gas

fuel a substance that will release energy when it reacts with another substance; in this context the other substance is usually oxygen

galvanise to galvanise iron or steel means to cover it with a thin layer of zinc; the zinc reacts with oxygen to form zinc oxide when it is exposed to air and this forms a strong and impenetrable coating

galvanised metal metal that is covered by thin layers of zinc and zinc oxide

generate to produce something; in this case it refers to some other source of energy being converted to electrical energy (electrical power or electricity)

group the vertical columns of the Periodic Table are called groups

identical exactly the same in every way

ignite to set something on fire

indicator a substance that changes colour in the presence of another substance, showing that that substance is present

inert unreactive; these substances do not react with other substances and do not change into other compounds

IUPAC International Union of Pure and Applied Chemistry (acronym)

IUPAC system a system for naming compounds in a way that is unique for each compound

laboratory acids acid commonly found in the laboratory

litmus a well-known acid-base indicator that turns red when mixed with an acid and blue when mixed with a base

macroscopic the macroscopic world includes all the things we can observe with our five senses – things we can see, hear, smell, touch and taste

metal an element that is shiny, ductile and malleable; metals occur on the left and towards the middle of the Periodic Table

metal carbonate a compound with the general formula MCO or M_2CO_3 where M represents a metal atom, C represents a carbon atom and O represents an oxygen atom

metal hydroxide a compound with the general formula MOH or $M(OH)_2$ where M represents a metal atom, O represents an oxygen atom and H represents a hydrogen atom

metal oxide the product of the reaction between a metal and oxygen; a compound with the general formula MO or M_2O where M represents a metal atom and O represents an oxygen atom

molecule two or more atoms that have chemically bonded with each other; the atoms in a molecule can be of the same kind (in which case it would be a molecule of an element), or they can be of different kinds (in which case it would be a molecule of a compound)

neutralisation reaction a reaction in which the reactants neutralise each other

neutralise to neutralise something means to take away its potency

neutral solution a solution with pH = 7

neutrons a type of sub-atomic particle similar to protons in mass and size, but neutral (without charge); neutrons together with protons make up the atomic nucleus

non-metal an element that does not have metallic properties; non-metals (excluding hydrogen) occur in the top right-hand corner of the Periodic Table

non-metal oxide the product of the reaction between a non-metal and oxygen

non-renewable non-renewable energy sources refer to sources that can be used up, such as fossil fuels (coal, oil, or natural gas)

oxidise when a compound reacts with oxygen, we say it is oxidised; in chemistry, the word oxidise means much more than this, but for now, we will limit ourselves to this simple definition

penetrate when liquid or air penetrates into a material, it passes into or through that material (usually because of tiny holes in the material); something that cannot be penetrated is called impenetrable

Periodic Table a table in which the chemical elements are arranged in order of increasing atomic number

period the horizontal rows of the Periodic Table are called periods

pharmacist a person who has studied pharmacy and uses this knowledge in the field of health science

pH pH measures the acidity and alkalinity of a solution as a number on a scale ranging between 0 and 14

picture equation an equation that describes a chemical reaction using diagrams of the particles of the compounds involved in the reaction

porous material that has tiny holes through which liquid or air may pass

potency power

prefix a bit added at the start of a word, usually to indicate number, e.g.,mono-, di-, or tri

presence the state of something existing or being present in a place

preservative a substance that is added to products (usually food or beverages) to make them last longer; most preservatives are toxic to microorganisms, but are added in such small quantities that they do not pose significant harm to humans

product a substance that forms during the reaction; it will be present after the reaction has taken place

protons sub-atomic particles that are positively charged and occur inside the atomic nucleus along with neutrons

reactant the starting substances that undergo change in a chemical reaction

reactive elements and compounds that are reactive will readily react with many other substances

red cabbage indicator An acid-base indicator made from the sap of red cabbage; red cabbage indicator is also capable of displaying a range of colours, depending on the pH of the solution with which it is mixed

renewable a renewable source of energy cannot be used up, such as water, wind, or solar power

rust a reddish or yellowish-brown, often flaky, coating of iron oxide that is formed on iron or steel by oxidation (when it reacts with oxygen in the air)

rust-resistant a rust-resistant material is one that does not rust

semi-metal an element that has properties of both metals and non-metals; the semi-metals occur in a narrow diagonal strip that separates the metals form the non-metals on the periodic table

steel a metal alloy composed of a mixture of iron and other elements (mostly metal); it is very strong and used widely in the construction industry (also in buildings)

submicroscopic the submicroscopic world includes things that exist but that we can't see; atoms and molecules can only be 'seen' as mental pictures and when we draw these, we refer to them as 'submicroscopic diagrams'

subscript a number that is placed inside a chemical formula; it shows the number of atoms of that type in one molecule of that compound, for example O_2

suffix a bit added at the end of a word, for example, -ide

symbol (or element symbol) a unique letter (or letters) that represent a given element

symbolic the symbolic world includes letters and numbers that we use to represent atoms and molecules

systematic name The unique name that will be generated for a given compound, if the IUPAC system for naming compounds is followed correctly

tarnish when a metal surface gets dirty or spotty after reacting with oxygen or other substances in the air, we say it is tarnished

toxic poisonous, harmful to living organisms

unique the only one of its kind; unlike anything else

unit a quantity used as a standard of measurement, for example, units of time are second, minute, hour, day, week, month, year and decade

universal indicator An acid-base indicator that can display a range of colours, depending on the pH of the solution with which it is mixed word equation an equation that describes a chemical reaction using the names of the compounds involved in the reaction